ADVANCED COMPUTER NETWORKS-PCS223

Submitted by - Anchal
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ME - CSE1

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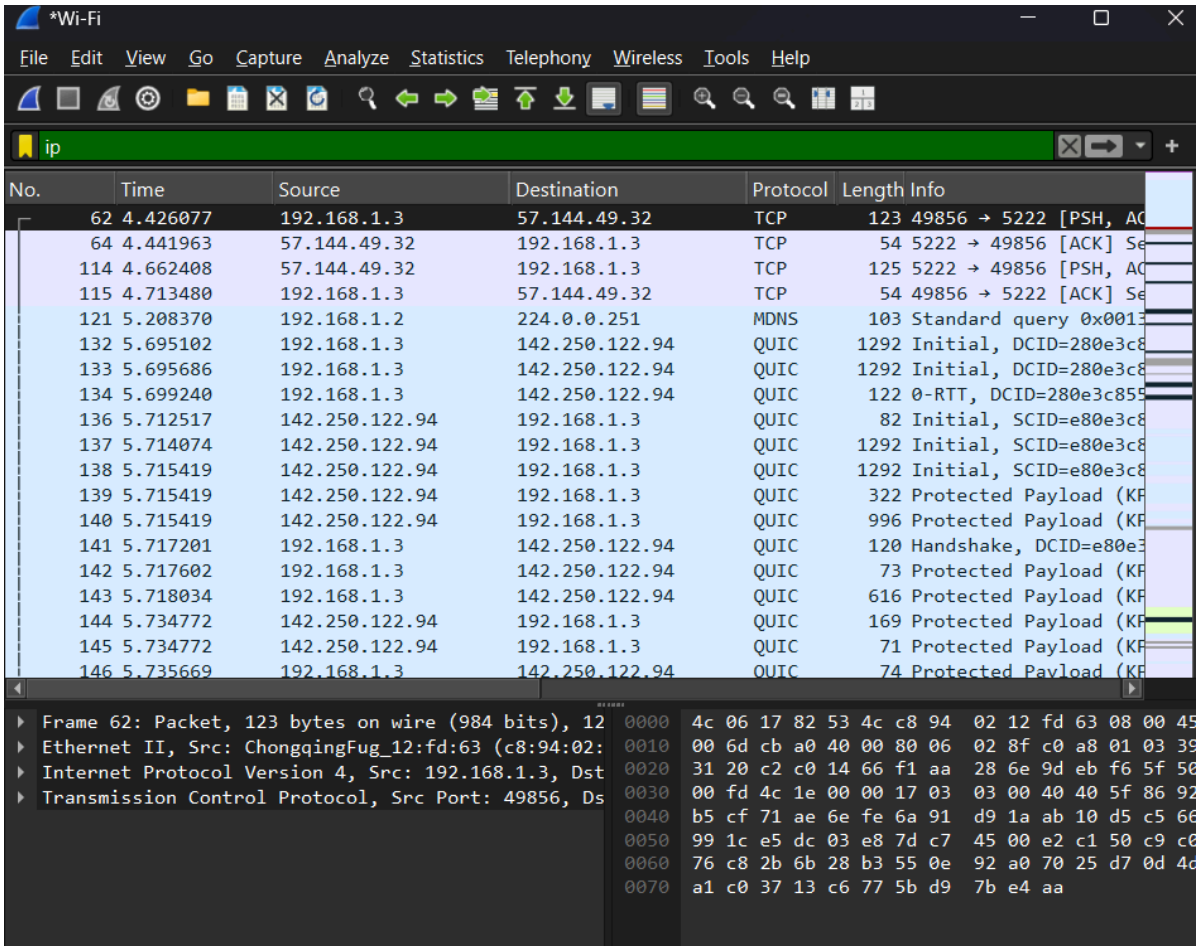
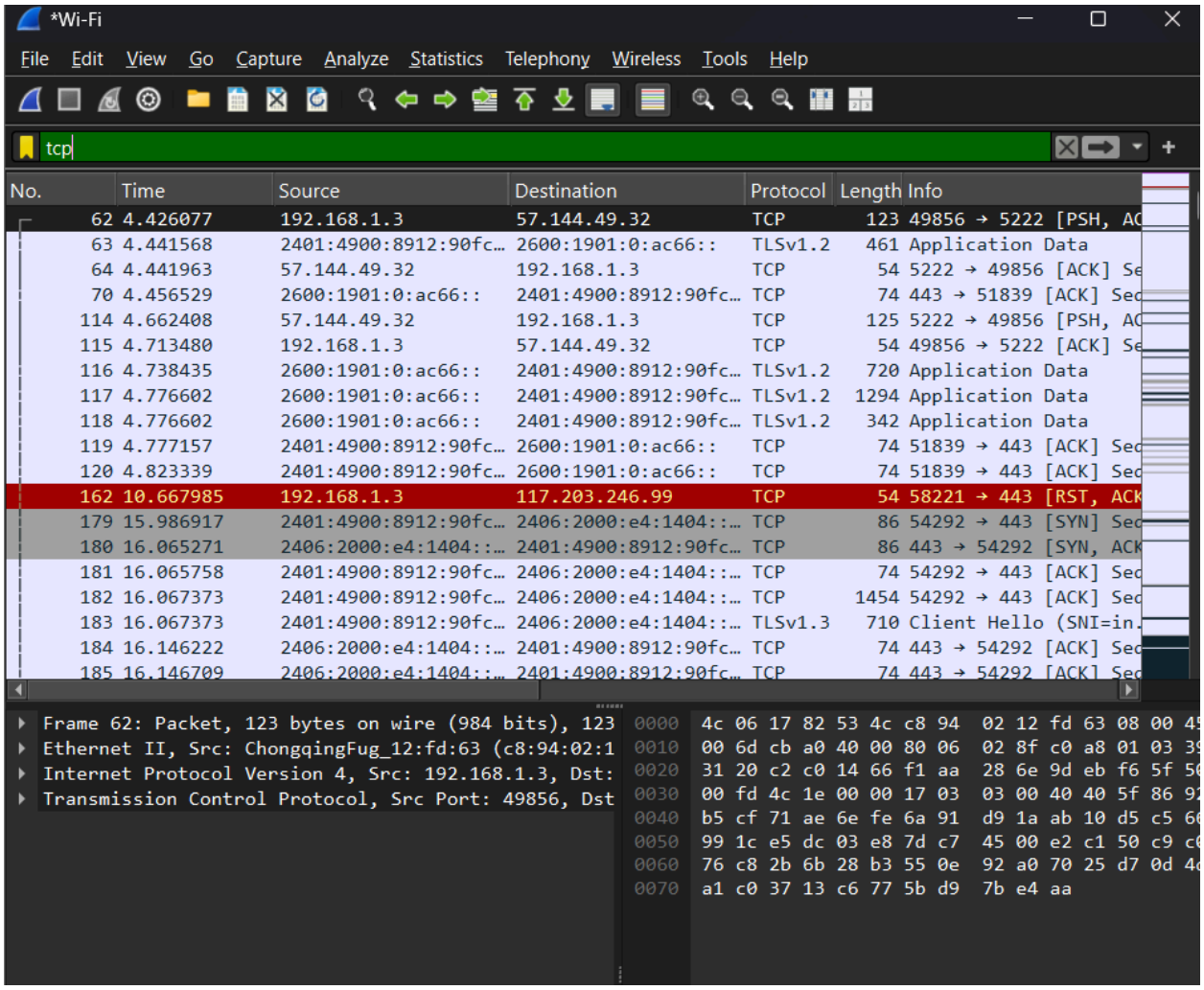
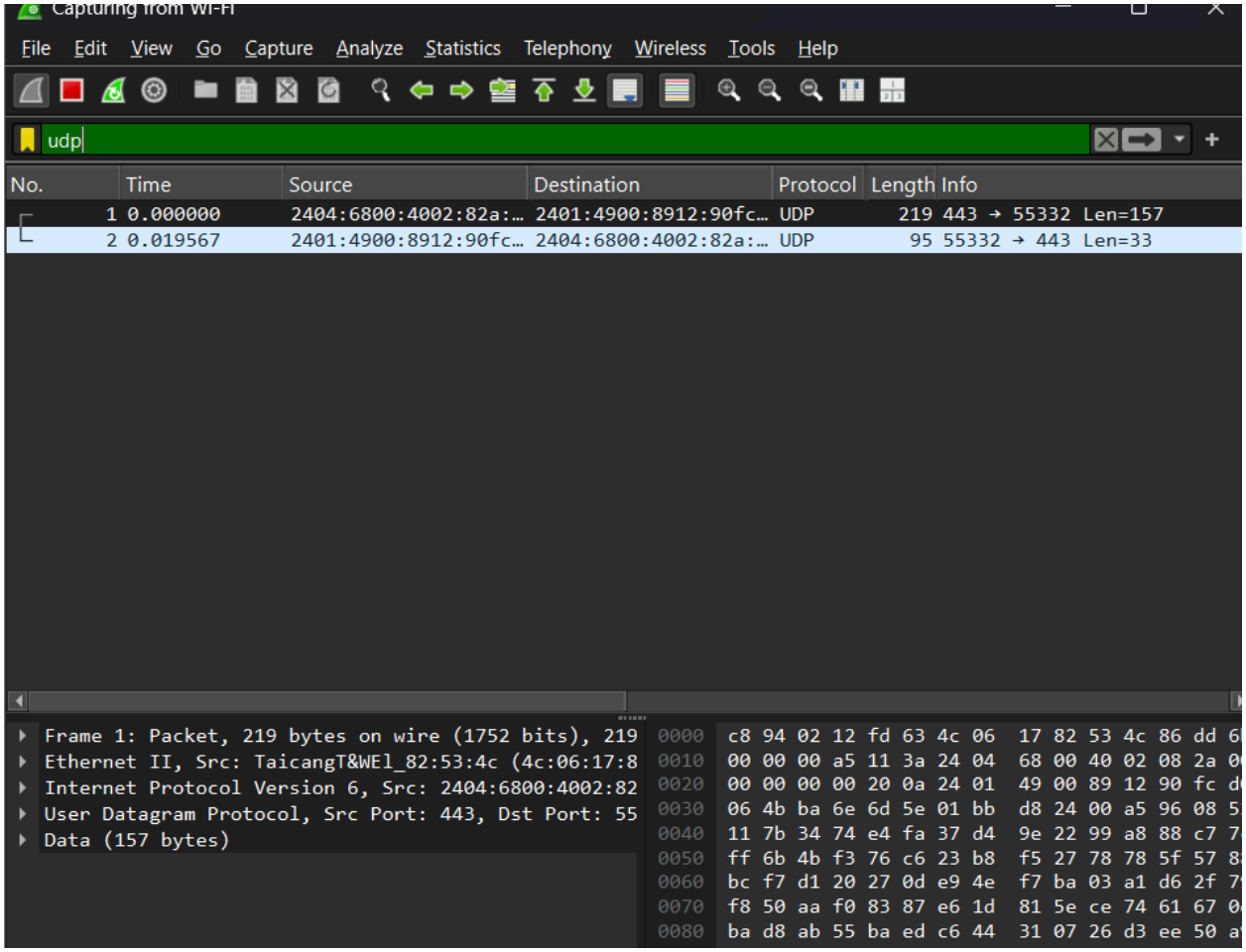
2026

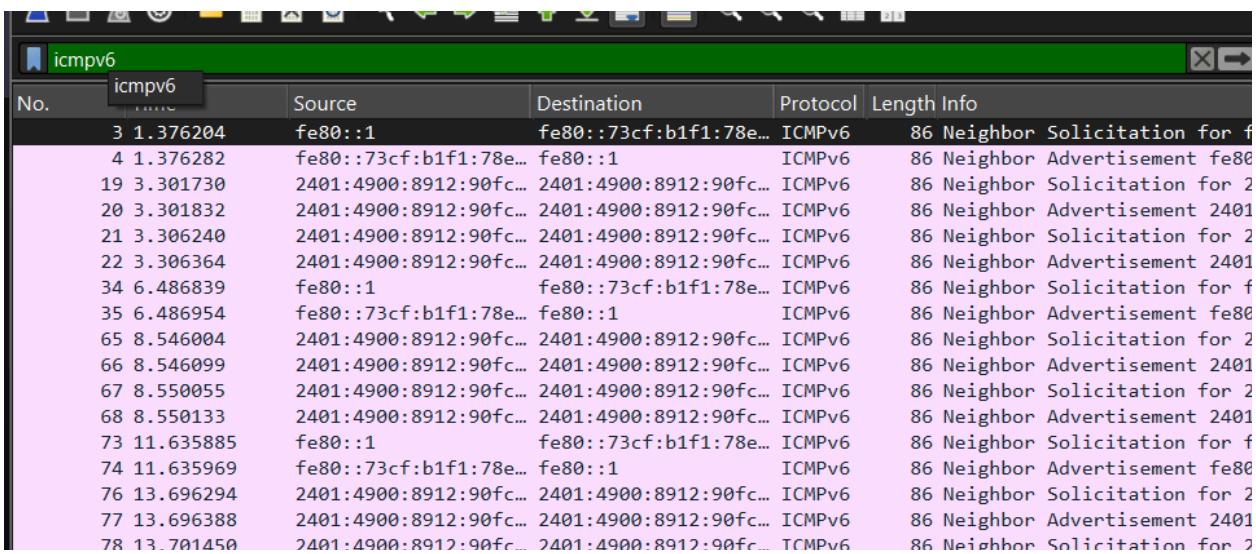
Assignment-1

1. To capture live network packets using Wireshark and identify basic network protocols such as TCP, UDP, IP, and DNS.

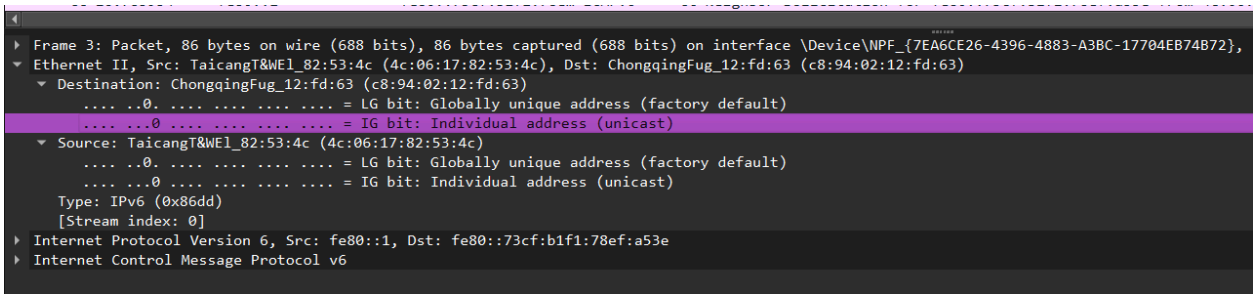



*Wi-Fi
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
tcp
No. Time Source Destination Protocol Length Info
62 4.426077 192.168.1.3 57.144.49.32 TCP 123 49856 → 5222 [PSH, AC
63 4.441568 2401:4900:8912:90fc… 2600:1901:0:ac66:: TLSv1.2 461 Application Data
64 4.441963 57.144.49.32 192.168.1.3 TCP 54 5222 → 49856 [ACK] Se
70 4.456529 2600:1901:0:ac66:: 2401:4900:8912:90fc… TCP 74 443 → 51839 [ACK] Sec
114 4.662408 57.144.49.32 192.168.1.3 TCP 125 5222 → 49856 [PSH, AC
115 4.713480 192.168.1.3 57.144.49.32 TCP 54 49856 → 5222 [ACK] Se
116 4.738435 2600:1901:0:ac66:: 2401:4900:8912:90fc… TLSv1.2 720 Application Data
117 4.776602 2600:1901:0:ac66:: 2401:4900:8912:90fc… TLSv1.2 1294 Application Data
118 4.776602 2600:1901:0:ac66:: 2401:4900:8912:90fc… TLSv1.2 342 Application Data
119 4.777157 2401:4900:8912:90fc… 2600:1901:0:ac66:: TCP 74 51839 → 443 [ACK] Sec
120 4.823339 2401:4900:8912:90fc… 2600:1901:0:ac66:: TCP 74 51839 → 443 [ACK] Sec
162 10.667985 192.168.1.3 117.203.246.99 TCP 54 58221 → 443 [RST, ACK
179 15.986917 2401:4900:8912:90fc… 2406:2000:e4:1404::… TCP 86 54292 → 443 [SYN] Sec
180 16.065271 2406:2000:e4:1404::… 2401:4900:8912:90fc… TCP 86 443 → 54292 [SYN, ACK
181 16.065758 2401:4900:8912:90fc… 2406:2000:e4:1404::… TCP 74 54292 → 443 [ACK] Sec
182 16.067373 2401:4900:8912:90fc… 2406:2000:e4:1404::… TCP 1454 54292 → 443 [ACK] Sec
183 16.067373 2401:4900:8912:90fc… 2406:2000:e4:1404::… TLSv1.3 710 Client Hello (SNI=in.
184 16.146222 2406:2000:e4:1404::… 2401:4900:8912:90fc… TCP 74 443 → 54292 [ACK] Sec
185 16.146709 2406:2000:e4:1404::… 2401:4900:8912:90fc… TCP 74 443 → 54292 [ACK] Sec
Frame 62: Packet, 123 bytes on wire (984 bits), 123
Ethernet II, Src: ChongqingFug_12:fd:63 (c8:94:02:1
Internet Protocol Version 4, Src: 192.168.1.3, Dst:
Transmission Control Protocol, Src Port: 49856, Dst
0000 4c 06 17 82 53 4c c8 94 02 12 fd 63 08 00 4
0010 00 6d cb a0 40 00 80 06 02 8f c0 a8 01 03 3
0020 31 20 c2 c0 14 66 f1 aa 28 6e 9d eb f6 5f 5
0030 00 fd 4c 1e 00 00 17 03 03 00 40 40 5f 86 9
0040 b5 cf 71 ae 6e fe 6a 91 d9 1a ab 10 d5 c5 6
0050 99 1c e5 dc 03 e8 7d c7 45 00 e2 c1 50 c9 c
0060 76 c8 2b 6b 28 b3 55 0e 92 a0 70 25 d7 0d 4
0070 a1 c0 37 13 c6 77 5b d9 7b e4 aa


Capturing from Wi-Fi
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
udp
No. Time Source Destination Protocol Length Info
1 0.000000 2404:6800:4002:82a:… 2401:4900:8912:90fc… UDP 219 443 → 55332 Len=157
2 0.019567 2401:4900:8912:90fc… 2404:6800:4002:82a:… UDP 95 55332 → 443 Len=33
Frame 1: Packet, 219 bytes on wire (1752 bits), 219
Ethernet II, Src: TaicangT&WEl_82:53:4c (4c:06:17:8
Internet Protocol Version 6, Src: 2404:6800:4002:82
User Datagram Protocol, Src Port: 443, Dst Port: 55
Data (157 bytes)
0000 c8 94 02 12 fd 63 4c 06 17 82 53 4c 86 dd
0010 00 00 00 a5 11 3a 24 04 68 00 40 02 08 2a
0020 00 00 00 00 20 0a 24 01 49 00 89 12 90 fc
0030 06 4b ba 6e 6d 5e 01 bb d8 24 00 a5 96 08
0040 11 7b 34 74 e4 fa 37 d4 9e 22 99 a8 88 c7
0050 ff 6b 4b f3 76 c6 23 b8 f5 27 78 78 5f 57
0060 bc f7 d1 20 27 0d e9 4e f7 ba 03 a1 d6 2f
0070 f8 50 aa f0 83 87 e6 1d 81 5e ce 74 61 67
0080 ba d8 ab 55 ba ed c6 44 31 07 26 d3 ee 50


icmpv6
icmpv6
No. Source Destination Protocol Length Info
3 1.376204 fe80::1 fe80::73cf:b1f1:78e… ICMPv6 86 Neighbor Solicitation for
4 1.376282 fe80::73cf:b1f1:78e… fe80::1 ICMPv6 86 Neighbor Advertisement fe80
19 3.301730 2401:4900:8912:90fc… 2401:4900:8912:90fc… ICMPv6 86 Neighbor Solicitation for 2
20 3.301832 2401:4900:8912:90fc… 2401:4900:8912:90fc… ICMPv6 86 Neighbor Advertisement 2401
21 3.306240 2401:4900:8912:90fc… 2401:4900:8912:90fc… ICMPv6 86 Neighbor Solicitation for 2
22 3.306364 2401:4900:8912:90fc… 2401:4900:8912:90fc… ICMPv6 86 Neighbor Advertisement 2401
34 6.486839 fe80::1 fe80::73cf:b1f1:78e… ICMPv6 86 Neighbor Solicitation for
35 6.486954 fe80::73cf:b1f1:78e… fe80::1 ICMPv6 86 Neighbor Advertisement fe80
65 8.546004 2401:4900:8912:90fc… 2401:4900:8912:90fc… ICMPv6 86 Neighbor Solicitation for 2
66 8.546099 2401:4900:8912:90fc… 2401:4900:8912:90fc… ICMPv6 86 Neighbor Advertisement 2401
67 8.550055 2401:4900:8912:90fc… 2401:4900:8912:90fc… ICMPv6 86 Neighbor Solicitation for 2
68 8.550133 2401:4900:8912:90fc… 2401:4900:8912:90fc… ICMPv6 86 Neighbor Advertisement 2401
73 11.635885 fe80::1 fe80::73cf:b1f1:78e… ICMPv6 86 Neighbor Solicitation for
74 11.635969 fe80::73cf:b1f1:78e… fe80::1 ICMPv6 86 Neighbor Advertisement fe80
76 13.696294 2401:4900:8912:90fc… 2401:4900:8912:90fc… ICMPv6 86 Neighbor Solicitation for 2
77 13.696388 2401:4900:8912:90fc… 2401:4900:8912:90fc… ICMPv6 86 Neighbor Advertisement 2401

OUTPUT-



| 1 | Frame Number | 3 |

| 2 | Packet Length | 86 bytes |

| 3 | Source MAC Address | 4c:06:17:82:53:4c |

| 4 | Destination MAC Address | c8:94:02:12:fd:63 |

| 5 | Source MAC Manufacturer | TaicangT&WE1 |

| 6 | Destination MAC Manufacturer | ChongqingFug |

| 7 | Ethernet Type | 0x86dd (IPv6) |

| 8 | IP Version | IPv6 |

| 9 | Source IP Address | fe80::1 |

| 10 | Destination IP Address | fe80::73cf:b1f1:78ef:a53e |

| 11 | IP Address Type | Link-local (fe80::/10) |

| 12 | Transport Protocol | ICMPv6 |

| 13 | LG Bit (Source) | 0 (Globally unique) |

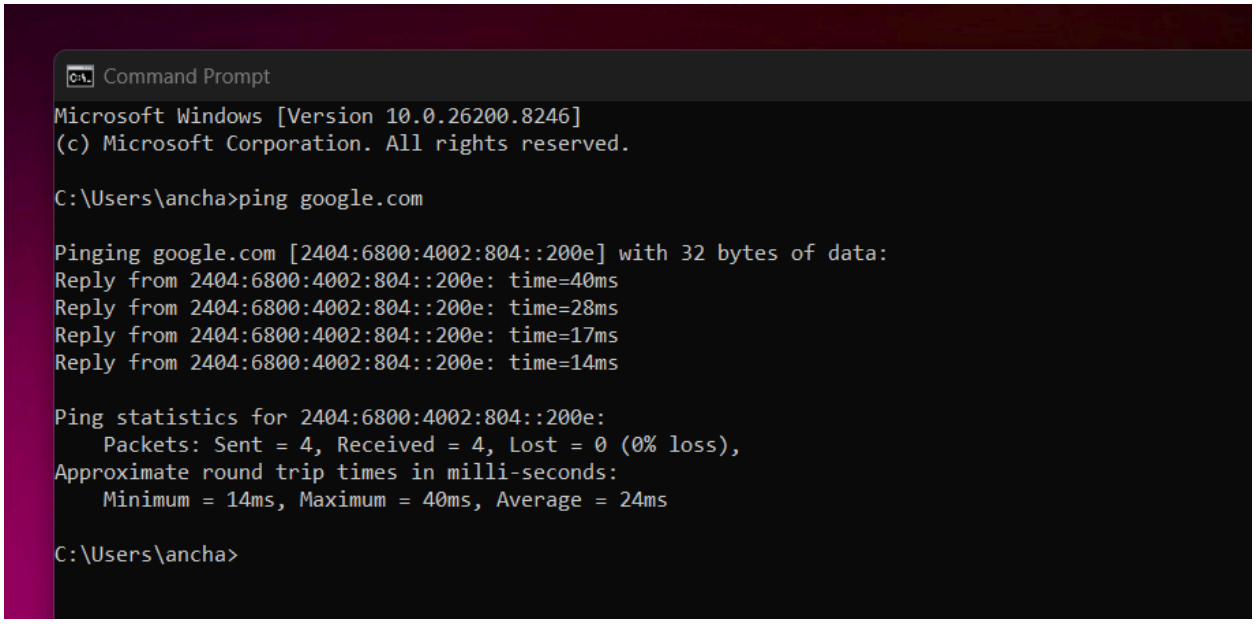
| 14 | IG Bit (Source) | 0 (Unicast) |

| 15 | Capture Interface | NPF_{7EA6CE26-4396-4883-A3BC-17704EB74B72} |

Assignment 2

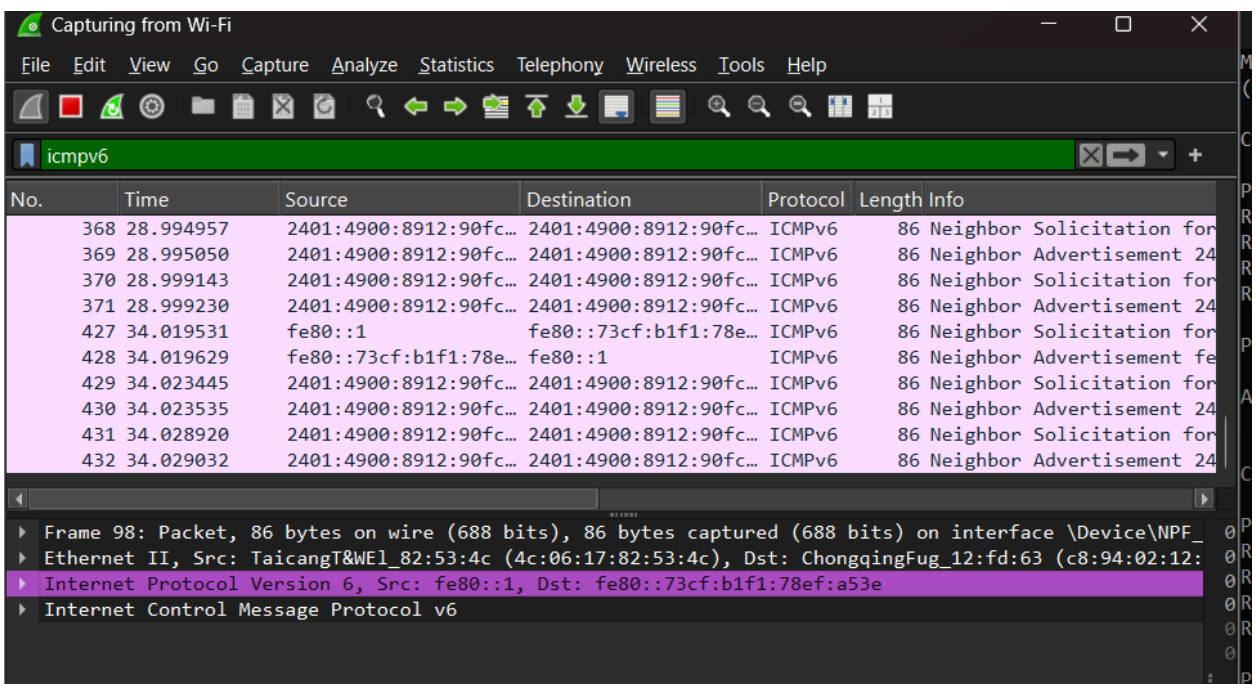
1. To capture and analyze ICMP packets (Ping request and reply) 2. To analyze TCP packets and observe the TCP three-way handshake 3. To capture and analyze UDP packets 4. To capture and analyze DNS query and response packets 5. To capture and analyze HTTP packets 6. To analyze ARP request and reply packets

Cmd-



```
Command Prompt
Microsoft Windows [Version 10.0.26200.8246]
(c) Microsoft Corporation. All rights reserved.

C:\Users\ancha>ping google.com

Pinging google.com [2404:6800:4002:804::200e] with 32 bytes of data:
Reply from 2404:6800:4002:804::200e: time=40ms
Reply from 2404:6800:4002:804::200e: time=28ms
Reply from 2404:6800:4002:804::200e: time=17ms
Reply from 2404:6800:4002:804::200e: time=14ms

Ping statistics for 2404:6800:4002:804::200e:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 14ms, Maximum = 40ms, Average = 24ms

C:\Users\ancha>
```

Wireshark - Via Wifi



Capturing from Wi-Fi

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

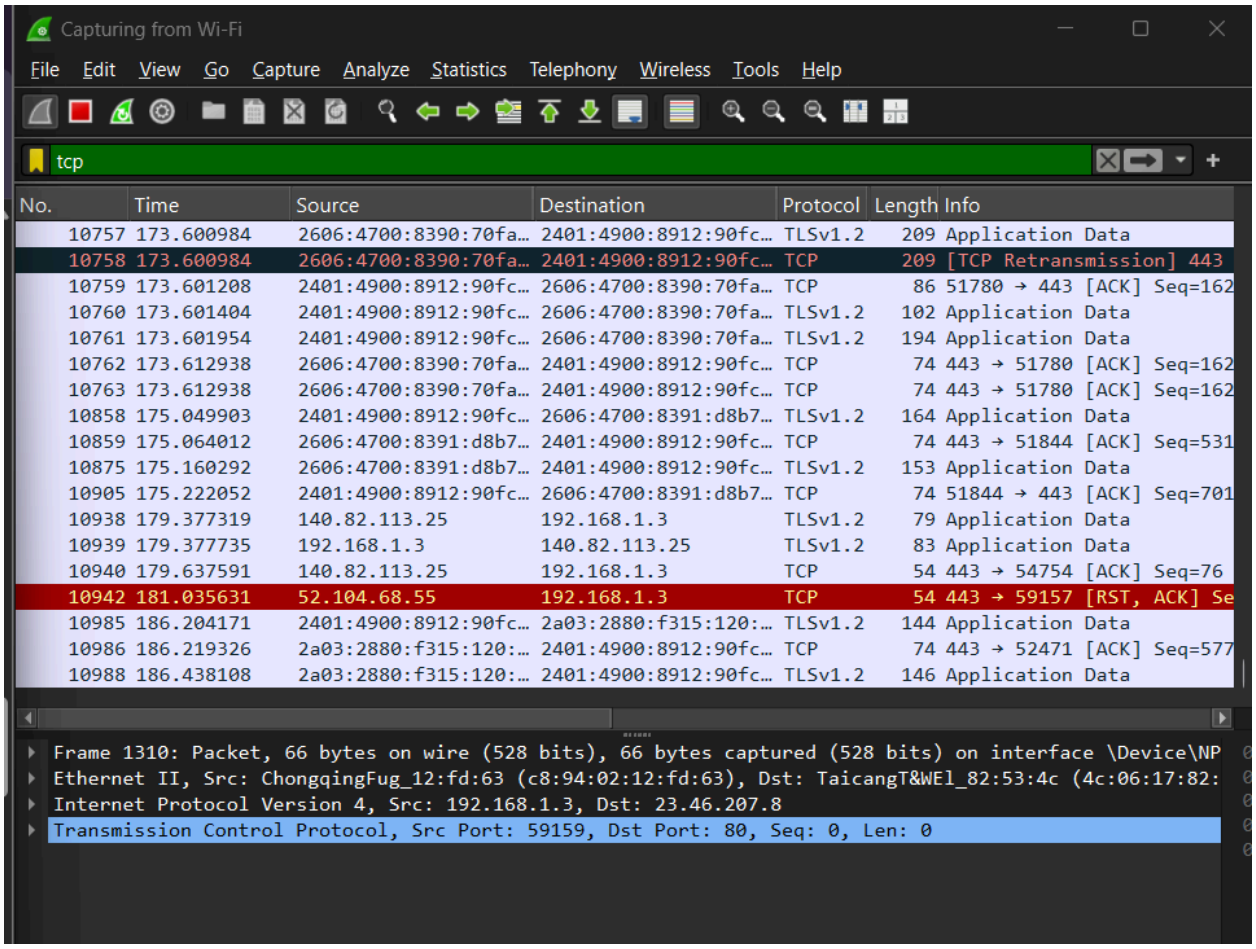
icmpv6

No.	Time	Source	Destination	Protocol	Length	Info
368	28.994957	2401:4900:8912:90fc…	2401:4900:8912:90fc…	ICMPv6	86	Neighbor Solicitation for
369	28.995050	2401:4900:8912:90fc…	2401:4900:8912:90fc…	ICMPv6	86	Neighbor Advertisement 24
370	28.999143	2401:4900:8912:90fc…	2401:4900:8912:90fc…	ICMPv6	86	Neighbor Solicitation for
371	28.999230	2401:4900:8912:90fc…	2401:4900:8912:90fc…	ICMPv6	86	Neighbor Advertisement 24
427	34.019531	fe80::1	fe80::73cf:b1f1:78e…	ICMPv6	86	Neighbor Solicitation for
428	34.019629	fe80::73cf:b1f1:78e…	fe80::1	ICMPv6	86	Neighbor Advertisement fe
429	34.023445	2401:4900:8912:90fc…	2401:4900:8912:90fc…	ICMPv6	86	Neighbor Solicitation for
430	34.023535	2401:4900:8912:90fc…	2401:4900:8912:90fc…	ICMPv6	86	Neighbor Advertisement 24
431	34.028920	2401:4900:8912:90fc…	2401:4900:8912:90fc…	ICMPv6	86	Neighbor Solicitation for
432	34.029032	2401:4900:8912:90fc…	2401:4900:8912:90fc…	ICMPv6	86	Neighbor Advertisement 24

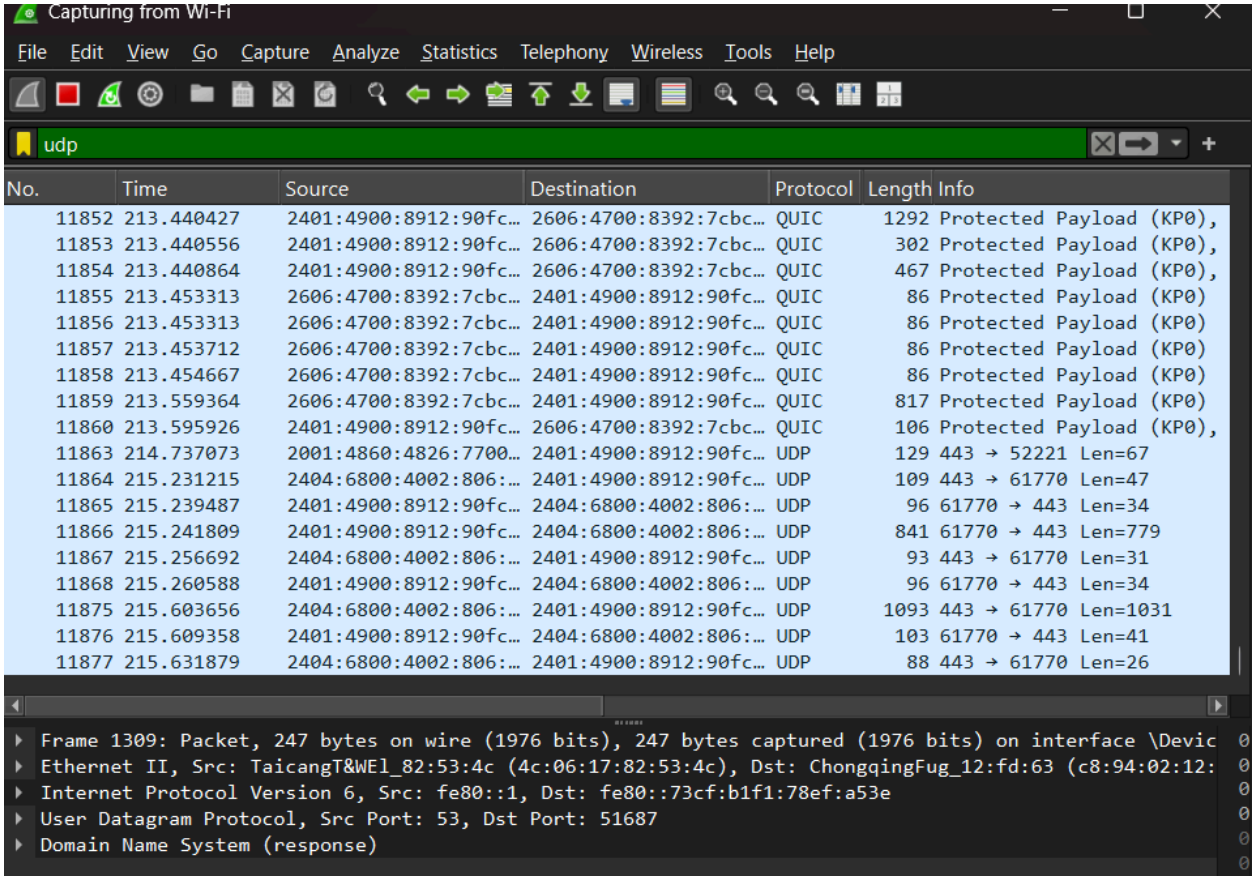
```
Frame 98: Packet, 86 bytes on wire (688 bits), 86 bytes captured (688 bits) on interface \Device\NPF_
Ethernet II, Src: TaicangT&WEl_82:53:4c (4c:06:17:82:53:4c), Dst: ChongqingFug_12:fd:63 (c8:94:02:12:
Internet Protocol Version 6, Src: fe80::1, Dst: fe80::73cf:b1f1:78ef:a53e
Internet Control Message Protocol v6
```

2.

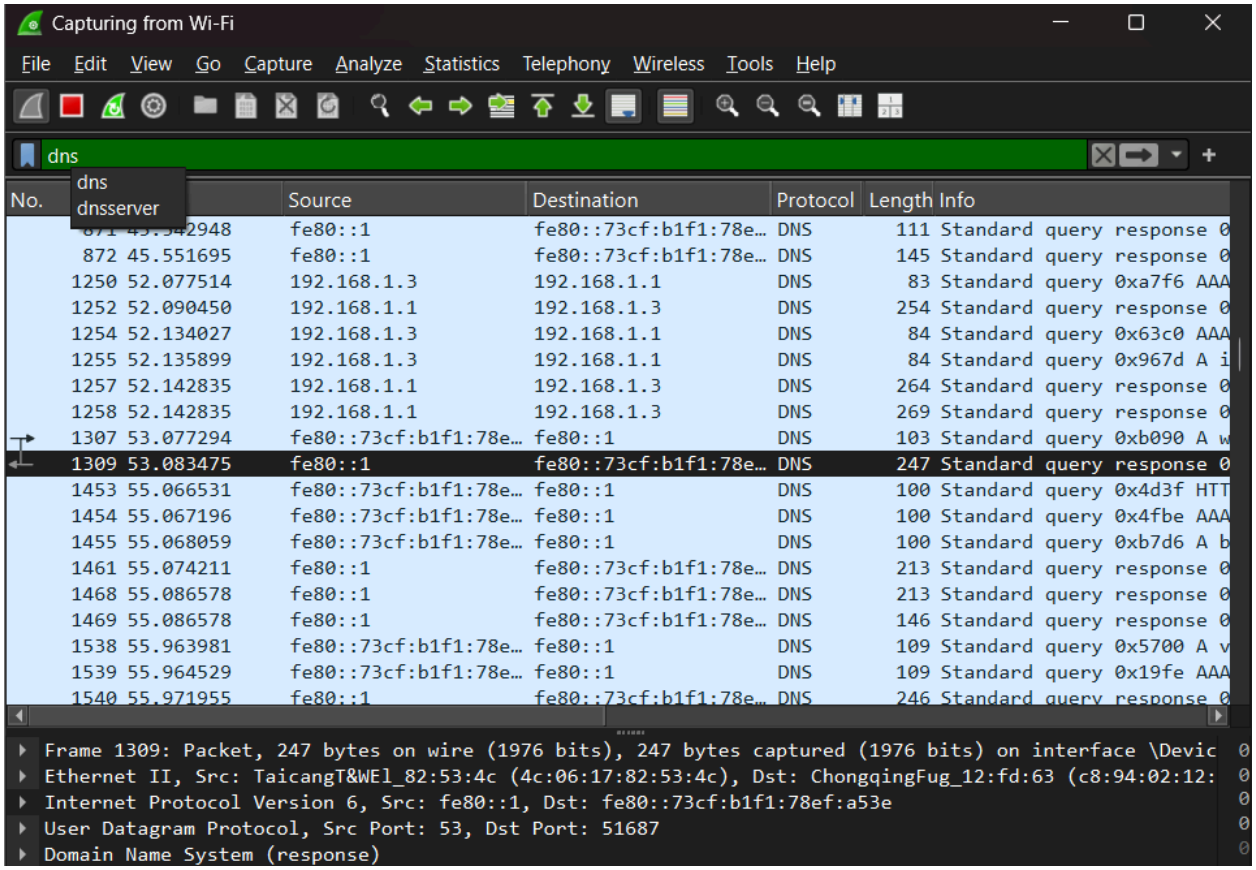
Tcp


Capturing from Wi-Fi
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
tcp
No. Time Source Destination Protocol Length Info
10757 173.600984 2606:4700:8390:70fa… 2401:4900:8912:90fc… TLSv1.2 209 Application Data
10758 173.600984 2606:4700:8390:70fa… 2401:4900:8912:90fc… TCP 209 [TCP Retransmission] 443
10759 173.601208 2401:4900:8912:90fc… 2606:4700:8390:70fa… TCP 86 51780 → 443 [ACK] Seq=162
10760 173.601404 2401:4900:8912:90fc… 2606:4700:8390:70fa… TLSv1.2 102 Application Data
10761 173.601954 2401:4900:8912:90fc… 2606:4700:8390:70fa… TLSv1.2 194 Application Data
10762 173.612938 2606:4700:8390:70fa… 2401:4900:8912:90fc… TCP 74 443 → 51780 [ACK] Seq=162
10763 173.612938 2606:4700:8390:70fa… 2401:4900:8912:90fc… TCP 74 443 → 51780 [ACK] Seq=162
10858 175.049903 2401:4900:8912:90fc… 2606:4700:8391:d8b7… TLSv1.2 164 Application Data
10859 175.064012 2606:4700:8391:d8b7… 2401:4900:8912:90fc… TCP 74 443 → 51844 [ACK] Seq=531
10875 175.160292 2606:4700:8391:d8b7… 2401:4900:8912:90fc… TLSv1.2 153 Application Data
10905 175.222052 2401:4900:8912:90fc… 2606:4700:8391:d8b7… TCP 74 51844 → 443 [ACK] Seq=701
10938 179.377319 140.82.113.25 192.168.1.3 TLSv1.2 79 Application Data
10939 179.377735 192.168.1.3 140.82.113.25 TLSv1.2 83 Application Data
10940 179.637591 140.82.113.25 192.168.1.3 TCP 54 443 → 54754 [ACK] Seq=76
10942 181.035631 52.104.68.55 192.168.1.3 TCP 54 443 → 59157 [RST, ACK] Se
10985 186.204171 2401:4900:8912:90fc… 2a03:2880:f315:120:… TLSv1.2 144 Application Data
10986 186.219326 2a03:2880:f315:120:… 2401:4900:8912:90fc… TCP 74 443 → 52471 [ACK] Seq=577
10988 186.438108 2a03:2880:f315:120:… 2401:4900:8912:90fc… TLSv1.2 146 Application Data
Frame 1310: Packet, 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface \Device\NP
Ethernet II, Src: ChongqingFug_12:fd:63 (c8:94:02:12:fd:63), Dst: TaicangT&WEl_82:53:4c (4c:06:17:82:
Internet Protocol Version 4, Src: 192.168.1.3, Dst: 23.46.207.8
Transmission Control Protocol, Src Port: 59159, Dst Port: 80, Seq: 0, Len: 0

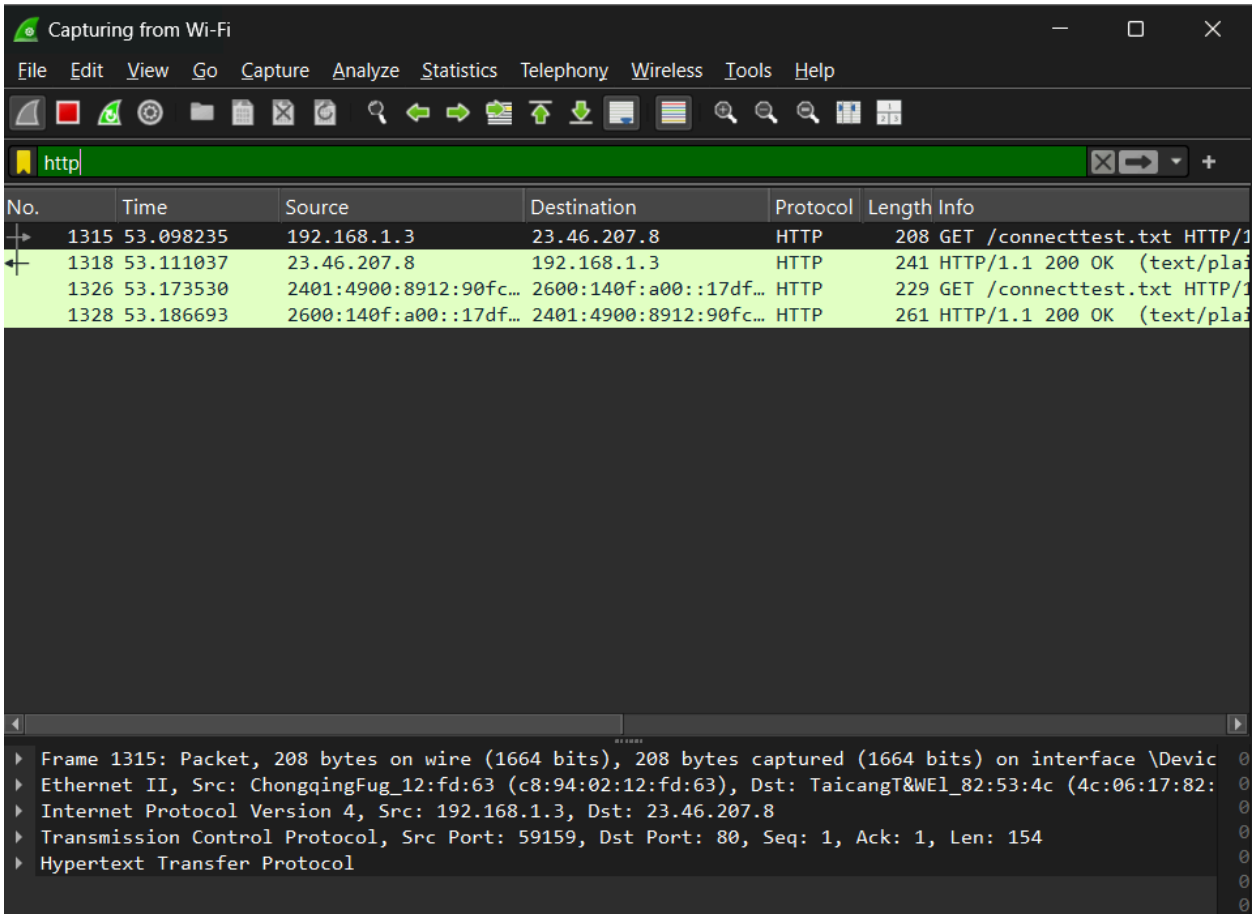
3.UDP


Capturing from Wi-Fi
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
udp
No. Time Source Destination Protocol Length Info
11852 213.440427 2401:4900:8912:90fc… 2606:4700:8392:7cbc… QUIC 1292 Protected Payload (KP0),
11853 213.440556 2401:4900:8912:90fc… 2606:4700:8392:7cbc… QUIC 302 Protected Payload (KP0),
11854 213.440864 2401:4900:8912:90fc… 2606:4700:8392:7cbc… QUIC 467 Protected Payload (KP0),
11855 213.453313 2606:4700:8392:7cbc… 2401:4900:8912:90fc… QUIC 86 Protected Payload (KP0)
11856 213.453313 2606:4700:8392:7cbc… 2401:4900:8912:90fc… QUIC 86 Protected Payload (KP0)
11857 213.453712 2606:4700:8392:7cbc… 2401:4900:8912:90fc… QUIC 86 Protected Payload (KP0)
11858 213.454667 2606:4700:8392:7cbc… 2401:4900:8912:90fc… QUIC 86 Protected Payload (KP0)
11859 213.559364 2606:4700:8392:7cbc… 2401:4900:8912:90fc… QUIC 817 Protected Payload (KP0)
11860 213.595926 2401:4900:8912:90fc… 2606:4700:8392:7cbc… QUIC 106 Protected Payload (KP0),
11863 214.737073 2001:4860:4826:7700… 2401:4900:8912:90fc… UDP 129 443 → 52221 Len=67
11864 215.231215 2404:6800:4002:806:… 2401:4900:8912:90fc… UDP 109 443 → 61770 Len=47
11865 215.239487 2401:4900:8912:90fc… 2404:6800:4002:806:… UDP 96 61770 → 443 Len=34
11866 215.241809 2401:4900:8912:90fc… 2404:6800:4002:806:… UDP 841 61770 → 443 Len=779
11867 215.256692 2404:6800:4002:806:… 2401:4900:8912:90fc… UDP 93 443 → 61770 Len=31
11868 215.260588 2401:4900:8912:90fc… 2404:6800:4002:806:… UDP 96 61770 → 443 Len=34
11875 215.603656 2404:6800:4002:806:… 2401:4900:8912:90fc… UDP 1093 443 → 61770 Len=1031
11876 215.609358 2401:4900:8912:90fc… 2404:6800:4002:806:… UDP 103 61770 → 443 Len=41
11877 215.631879 2404:6800:4002:806:… 2401:4900:8912:90fc… UDP 88 443 → 61770 Len=26
Frame 1309: Packet, 247 bytes on wire (1976 bits), 247 bytes captured (1976 bits) on interface \Devic
Ethernet II, Src: TaicangT&WEl_82:53:4c (4c:06:17:82:53:4c), Dst: ChongqingFug_12:fd:63 (c8:94:02:12:
Internet Protocol Version 6, Src: fe80::1, Dst: fe80::73cf:b1f1:78ef:a53e
User Datagram Protocol, Src Port: 53, Dst Port: 51687
Domain Name System (response)

Dns -


Capturing from Wi-Fi
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
dns
dns
dnsserver
No. Source Destination Protocol Length Info
872 45.551695 fe80::1 fe80::73cf:b1f1:78e… DNS 145 Standard query response 0
1250 52.077514 192.168.1.3 192.168.1.1 DNS 83 Standard query 0xa7f6 AAA
1252 52.090450 192.168.1.1 192.168.1.3 DNS 254 Standard query response 0
1254 52.134027 192.168.1.3 192.168.1.1 DNS 84 Standard query 0x63c0 AAA
1255 52.135899 192.168.1.3 192.168.1.1 DNS 84 Standard query 0x967d A i
1257 52.142835 192.168.1.1 192.168.1.3 DNS 264 Standard query response 0
1258 52.142835 192.168.1.1 192.168.1.3 DNS 269 Standard query response 0
1307 53.077294 fe80::73cf:b1f1:78e… fe80::1 DNS 103 Standard query 0xb090 A w
1309 53.083475 fe80::1 fe80::73cf:b1f1:78e… DNS 247 Standard query response 0
1453 55.066531 fe80::73cf:b1f1:78e… fe80::1 DNS 100 Standard query 0x4d3f HTT
1454 55.067196 fe80::73cf:b1f1:78e… fe80::1 DNS 100 Standard query 0x4fbe AAA
1455 55.068059 fe80::73cf:b1f1:78e… fe80::1 DNS 100 Standard query 0xb7d6 A b
1461 55.074211 fe80::1 fe80::73cf:b1f1:78e… DNS 213 Standard query response 0
1468 55.086578 fe80::1 fe80::73cf:b1f1:78e… DNS 213 Standard query response 0
1469 55.086578 fe80::1 fe80::73cf:b1f1:78e… DNS 146 Standard query response 0
1538 55.963981 fe80::73cf:b1f1:78e… fe80::1 DNS 109 Standard query 0x5700 A v
1539 55.964529 fe80::73cf:b1f1:78e… fe80::1 DNS 109 Standard query 0x19fe AAA
Frame 1309: Packet, 247 bytes on wire (1976 bits), 247 bytes captured (1976 bits) on interface \Devic
Ethernet II, Src: TaicangT&WEl_82:53:4c (4c:06:17:82:53:4c), Dst: ChongqingFug_12:fd:63 (c8:94:02:12:
Internet Protocol Version 6, Src: fe80::1, Dst: fe80::73cf:b1f1:78ef:a53e
User Datagram Protocol, Src Port: 53, Dst Port: 51687
Domain Name System (response)

Http


Capturing from Wi-Fi
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
http
No. Time Source Destination Protocol Length Info
1315 53.098235 192.168.1.3 23.46.207.8 HTTP 208 GET /connecttest.txt HTTP/1
1318 53.111037 23.46.207.8 192.168.1.3 HTTP 241 HTTP/1.1 200 OK (text/plai
1326 53.173530 2401:4900:8912:90fc… 2600:140f:a00::17df… HTTP 229 GET /connecttest.txt HTTP/1
1328 53.186693 2600:140f:a00::17df… 2401:4900:8912:90fc… HTTP 261 HTTP/1.1 200 OK (text/plai
Frame 1315: Packet, 208 bytes on wire (1664 bits), 208 bytes captured (1664 bits) on interface \Devic
Ethernet II, Src: ChongqingFug_12:fd:63 (c8:94:02:12:fd:63), Dst: TaicangT&WEl_82:53:4c (4c:06:17:82:
Internet Protocol Version 4, Src: 192.168.1.3, Dst: 23.46.207.8
Transmission Control Protocol, Src Port: 59159, Dst Port: 80, Seq: 1, Ack: 1, Len: 154
Hypertext Transfer Protocol

ARp

arp

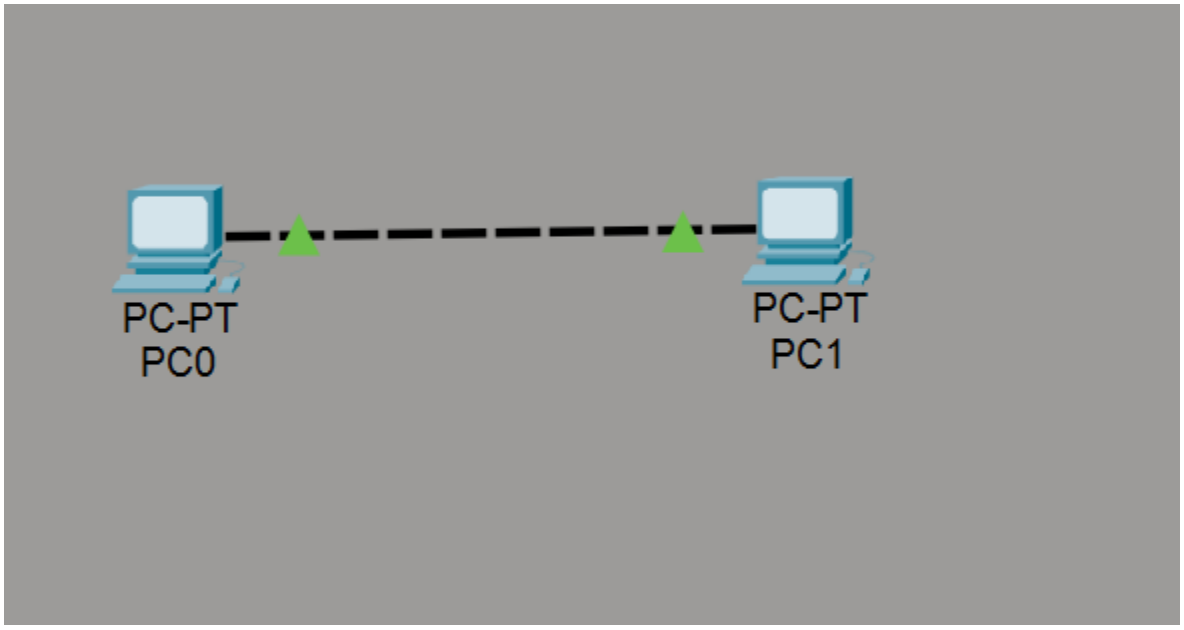
No.	Time	Source	Destination	Protocol	Length	Info
14364	325.046482	TaicangT&WEl_82:53:…	ChongqingFug_12:fd:…	ARP	60	Who has 192.168.1.3? Tel
14365	325.046501	ChongqingFug_12:fd:…	TaicangT&WEl_82:53:…	ARP	42	192.168.1.3 is at c8:94:
14449	330.162927	TaicangT&WEl_82:53:…	ChongqingFug_12:fd:…	ARP	60	Who has 192.168.1.3? Tel
14450	330.162956	ChongqingFug_12:fd:…	TaicangT&WEl_82:53:…	ARP	42	192.168.1.3 is at c8:94:
14537	335.362047	TaicangT&WEl_82:53:…	ChongqingFug_12:fd:…	ARP	60	Who has 192.168.1.3? Tel
14538	335.362071	ChongqingFug_12:fd:…	TaicangT&WEl_82:53:…	ARP	42	192.168.1.3 is at c8:94:
14599	340.256540	TaicangT&WEl_82:53:…	ChongqingFug_12:fd:…	ARP	60	Who has 192.168.1.3? Tel
14600	340.256594	ChongqingFug_12:fd:…	TaicangT&WEl_82:53:…	ARP	42	192.168.1.3 is at c8:94:
14697	345.678398	TaicangT&WEl_82:53:…	ChongqingFug_12:fd:…	ARP	60	Who has 192.168.1.3? Tel
14698	345.678428	ChongqingFug_12:fd:…	TaicangT&WEl_82:53:…	ARP	42	192.168.1.3 is at c8:94:
14712	350.806909	TaicangT&WEl_82:53:…	ChongqingFug_12:fd:…	ARP	60	Who has 192.168.1.3? Tel
14713	350.806943	ChongqingFug_12:fd:…	TaicangT&WEl_82:53:…	ARP	42	192.168.1.3 is at c8:94:
14757	354.935177	TaicangT&WEl_82:53:…	ChongqingFug_12:fd:…	ARP	60	Who has 192.168.1.3? Tel
14758	354.935250	ChongqingFug_12:fd:…	TaicangT&WEl_82:53:…	ARP	42	192.168.1.3 is at c8:94:
14874	360.099860	TaicangT&WEl_82:53:…	ChongqingFug_12:fd:…	ARP	60	Who has 192.168.1.3? Tel
14875	360.099880	ChongqingFug_12:fd:…	TaicangT&WEl_82:53:…	ARP	42	192.168.1.3 is at c8:94:
14974	365.212646	TaicangT&WEl_82:53:…	ChongqingFug_12:fd:…	ARP	60	Who has 192.168.1.3? Tel
14975	365.212676	ChongqingFug_12:fd:…	TaicangT&WEl_82:53:…	ARP	42	192.168.1.3 is at c8:94:

```
▸ Frame 1260: Packet, 42 bytes on wire (336 bits), 42 bytes captured (336 bits) on interface \Device\NP
▸ Ethernet II, Src: ChongqingFug_12:fd:63 (c8:94:02:12:fd:63), Dst: Broadcast (ff:ff:ff:ff:ff:ff)
▸ Address Resolution Protocol (ARP Probe)
```

ASSIGNMENT 3

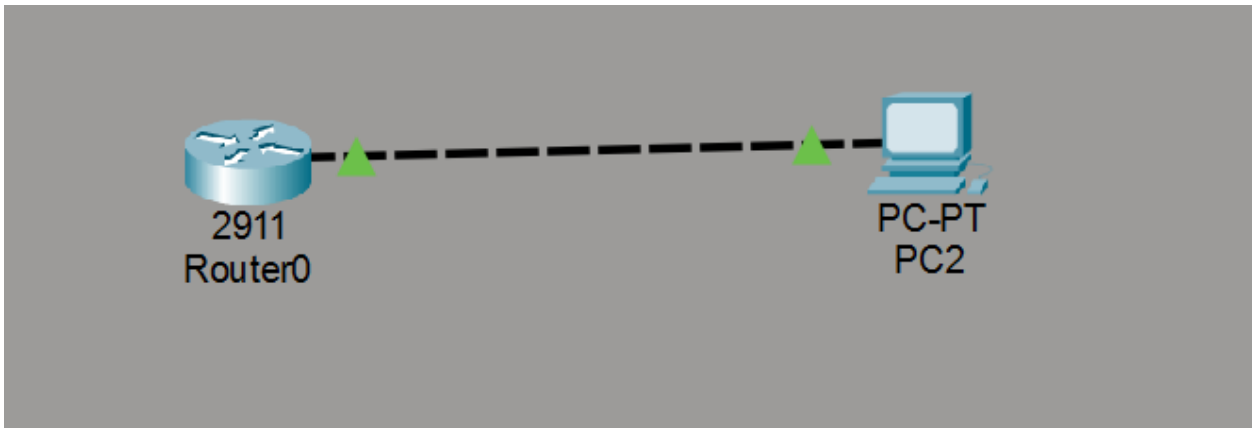
Ques1:

Sol-



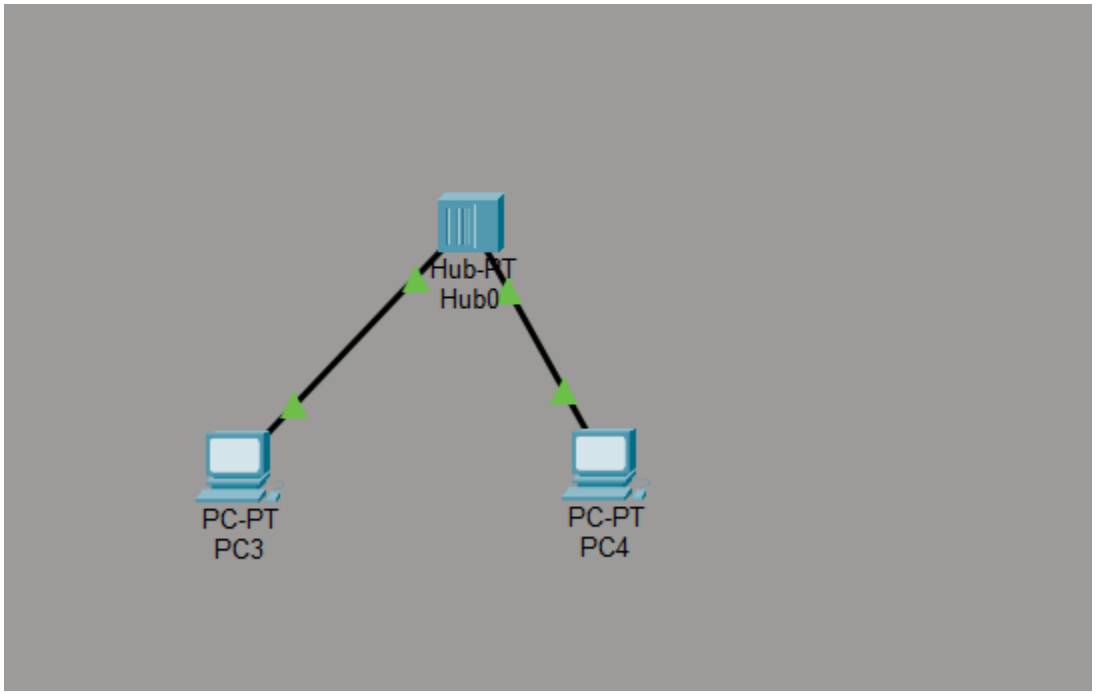
Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	PC0	PC1	ICMP		0.000	N	0	(edit)	(delete)

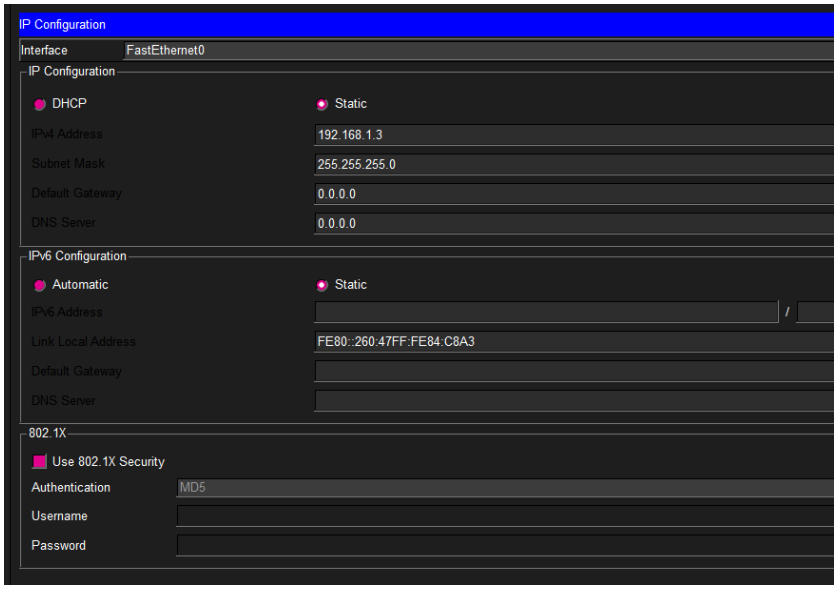
Ques 2:



Successful	PC2	Router0	ICMP		0.000	N	1	(edit)	(delete)

Ques 3:


Hub-PT
Hub0
PC-PT
PC3
PC-PT
PC4

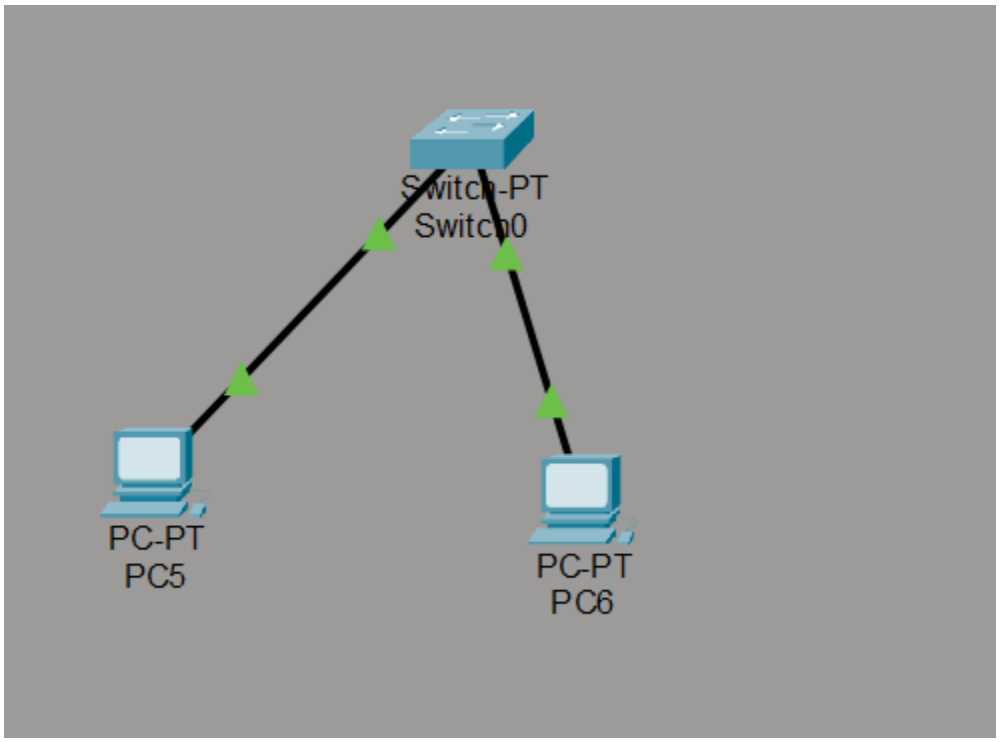

IP Configuration
Interface FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address 192.168.1.3
Subnet Mask 255.255.255.0
Default Gateway 0.0.0.0
DNS Server 0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address FE80::260:47FF:FE84:C8A3
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication MD5
Username
Password

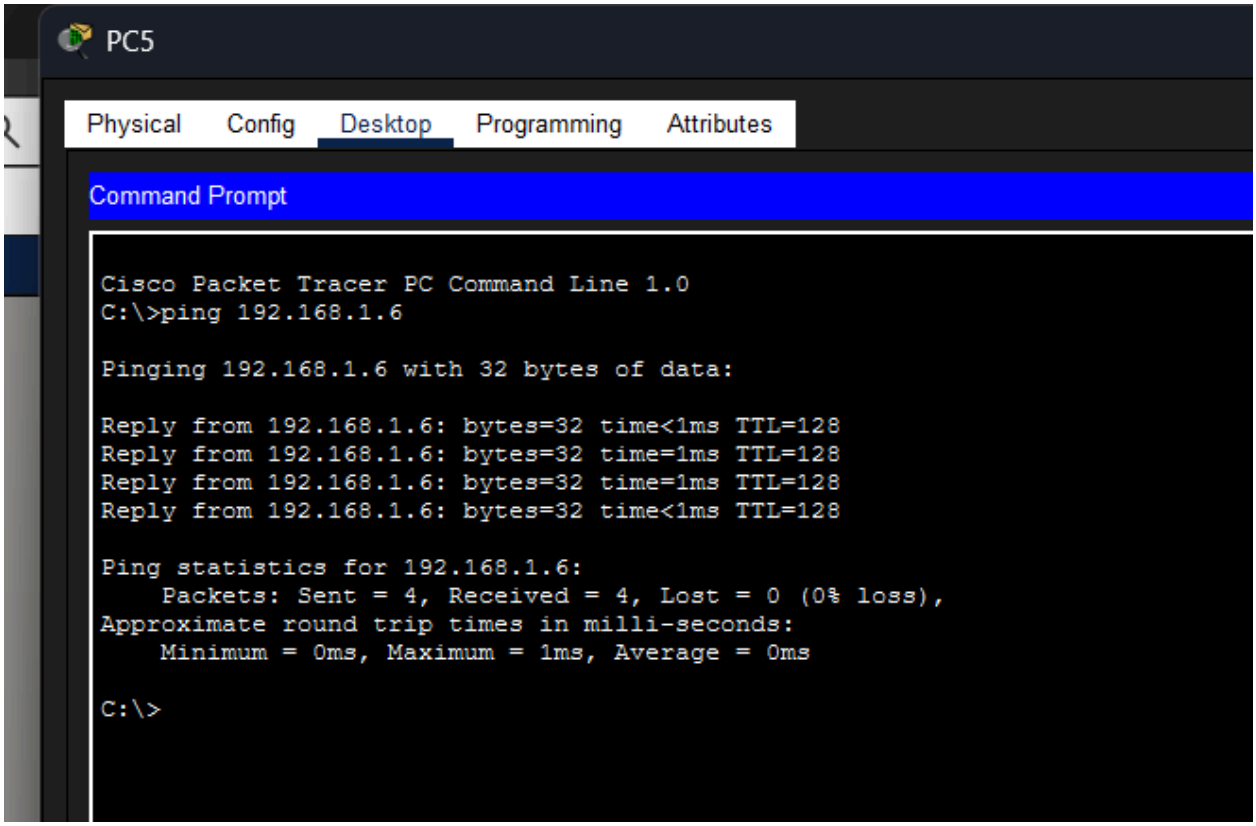
Successful PC3 PC4 ICMP 0.000 N 4 (edit) (delete)

```
C:\>ping 192.168.1.4

Pinging 192.168.1.4 with 32 bytes of data:

Reply from 192.168.1.4: bytes=32 time=1ms TTL=128
Reply from 192.168.1.4: bytes=32 time<1ms TTL=128
Reply from 192.168.1.4: bytes=32 time<1ms TTL=128
Reply from 192.168.1.4: bytes=32 time=1ms TTL=128

Ping statistics for 192.168.1.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
```

\

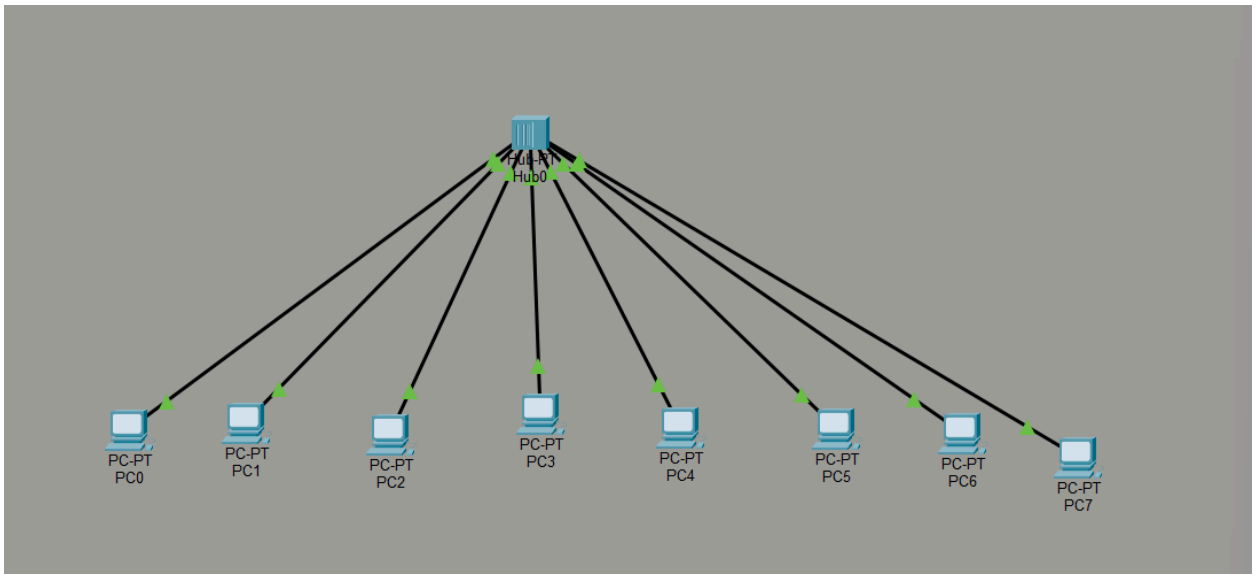
Ques 4:




PC5
Physical
Config
Desktop
Programming
Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.6
Pinging 192.168.1.6 with 32 bytes of data:
Reply from 192.168.1.6: bytes=32 time<1ms TTL=128
Reply from 192.168.1.6: bytes=32 time=1ms TTL=128
Reply from 192.168.1.6: bytes=32 time=1ms TTL=128
Reply from 192.168.1.6: bytes=32 time<1ms TTL=128
Ping statistics for 192.168.1.6:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 1ms, Average = 0ms
C:\>

Ques 5:

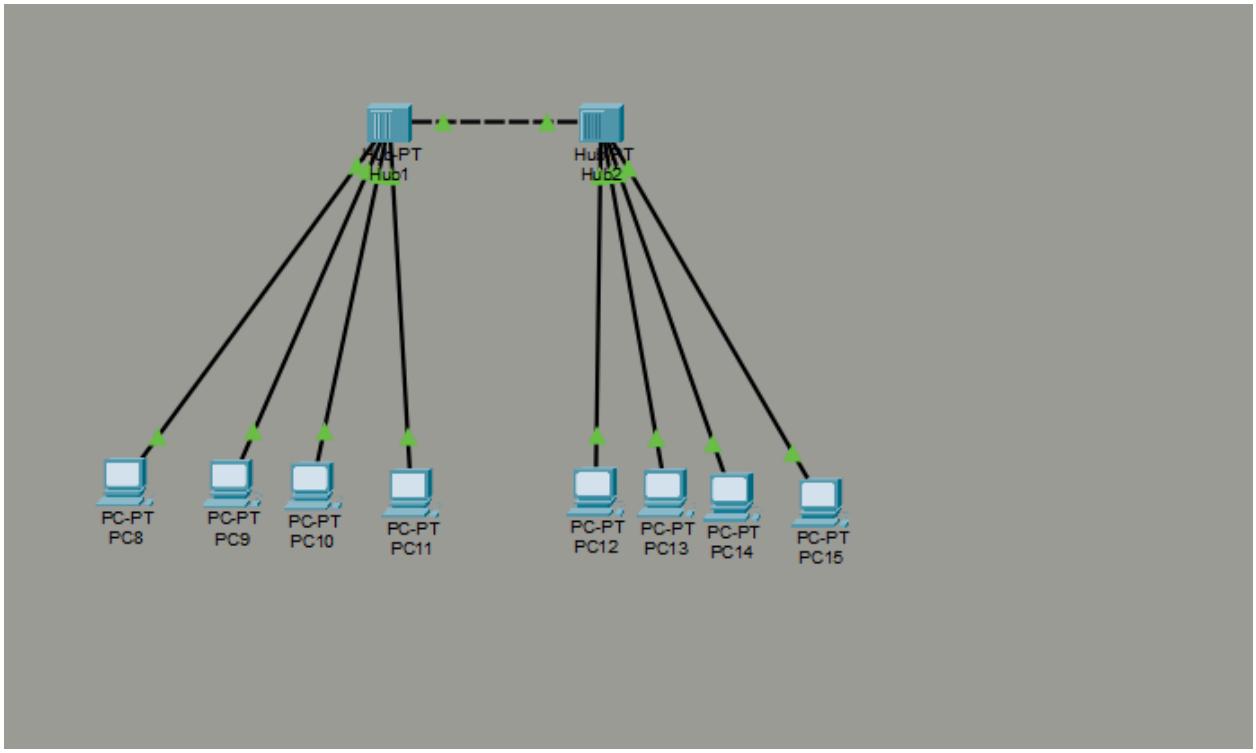
5(a)


Hub-PT
Hub0
PC-PT
PC0
PC-PT
PC1
PC-PT
PC2
PC-PT
PC3
PC-PT
PC4
PC-PT
PC5
PC-PT
PC6
PC-PT
PC7

Realtime Simulation

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	PC1	PC2	ICMP		0.000	N	4	(edit)	(delete)
	Successful	PC2	PC3	ICMP		0.000	N	5	(edit)	(delete)
	Successful	PC3	PC4	ICMP		0.000	N	6	(edit)	(delete)
	Successful	PC4	PC5	ICMP		0.000	N	7	(edit)	(delete)

5(b)



Realtime

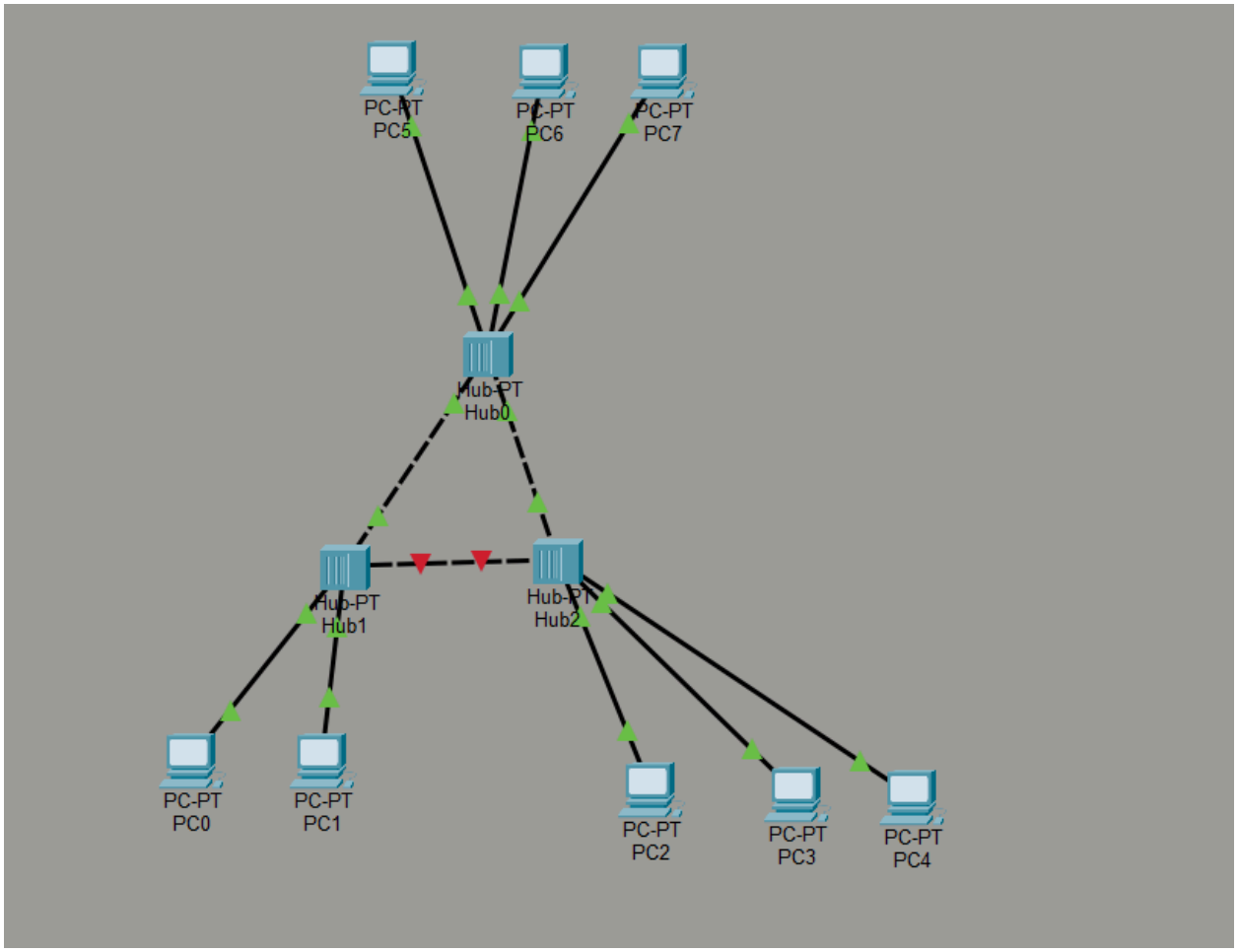
ario 0

Delete

ist Window

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	PC3	PC4	ICMP		0.000	N	6	(edit)	(delete)
	Successful	PC4	PC5	ICMP		0.000	N	7	(edit)	(delete)
	Successful	PC5	PC6	ICMP		0.000	N	8	(edit)	(delete)
	Successful	PC6	PC7	ICMP		0.000	N	9	(edit)	(delete)

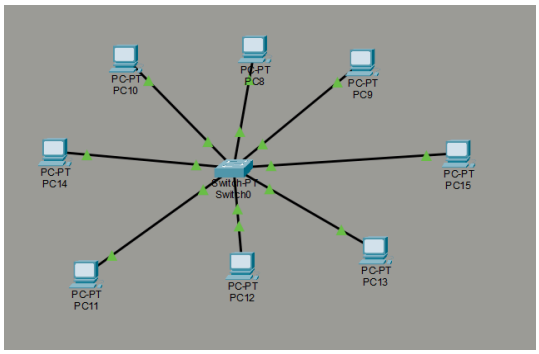
6.



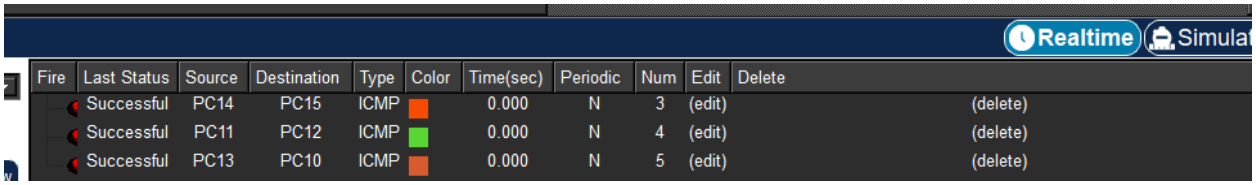
Realtime Simulati

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	PC1	PC7	ICMP		0.000	N	0	(edit)	(delete)
	Successful	PC1	PC2	ICMP		0.000	N	1	(edit)	(delete)
	Successful	PC0	PC1	ICMP		0.000	N	2	(edit)	(delete)

7.

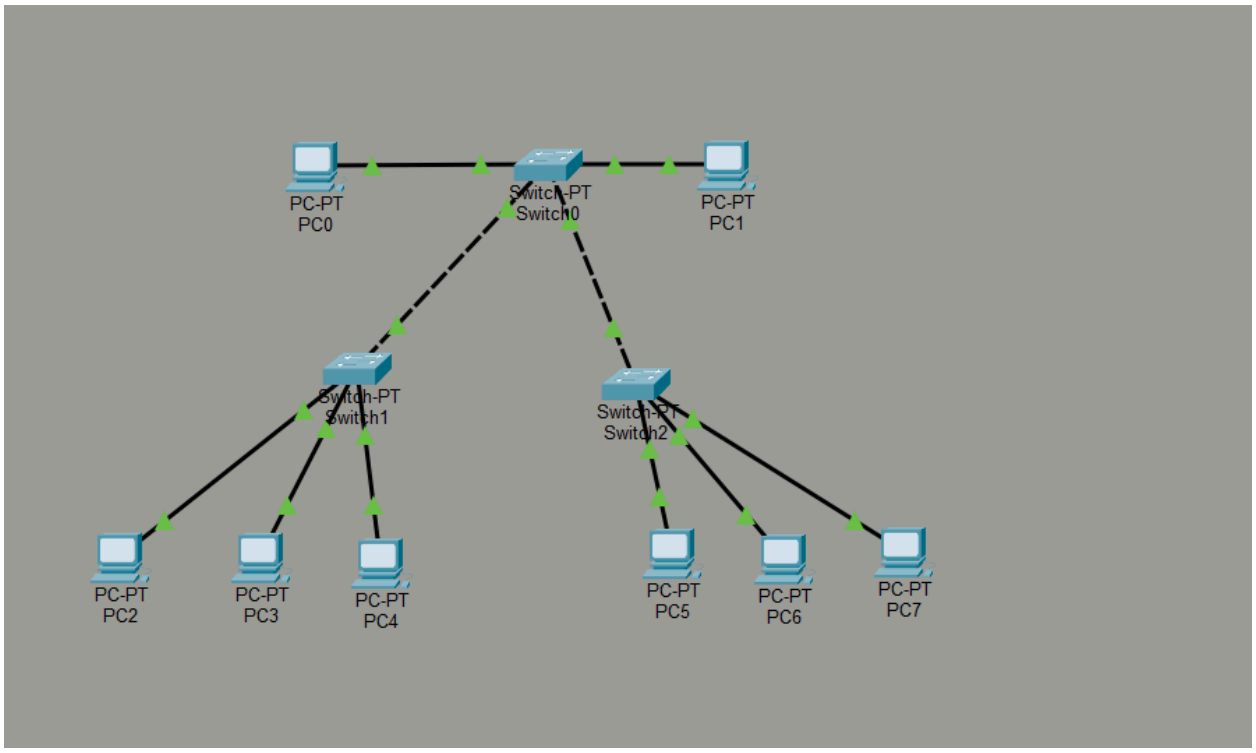


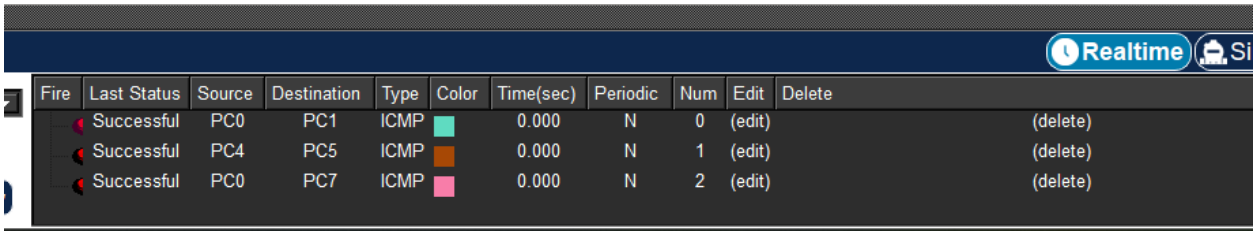
`



Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	PC14	PC15	ICMP		0.000	N	3	(edit)	(delete)
	Successful	PC11	PC12	ICMP		0.000	N	4	(edit)	(delete)
	Successful	PC13	PC10	ICMP		0.000	N	5	(edit)	(delete)

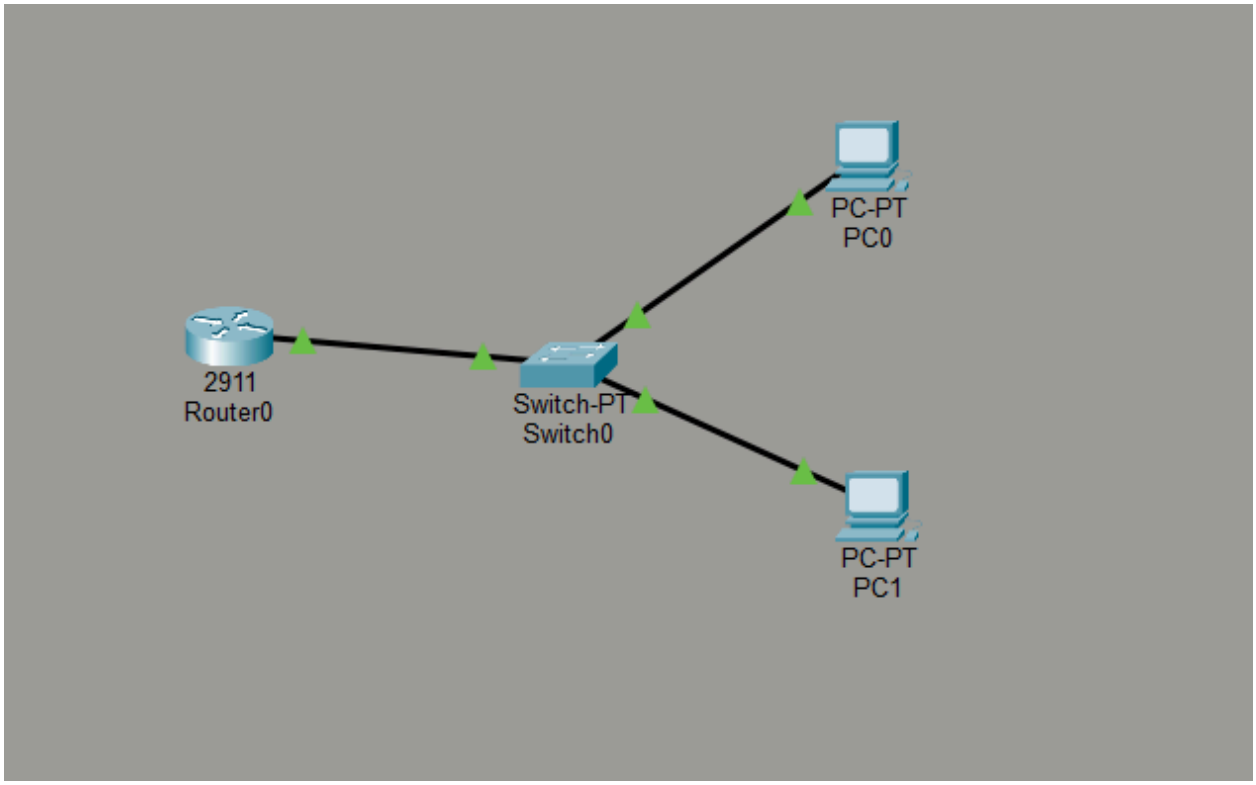
8.





Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	PC0	PC1	ICMP		0.000	N	0	(edit)	(delete)
	Successful	PC4	PC5	ICMP		0.000	N	1	(edit)	(delete)
	Successful	PC0	PC7	ICMP		0.000	N	2	(edit)	(delete)

ASSIGNMENT 4

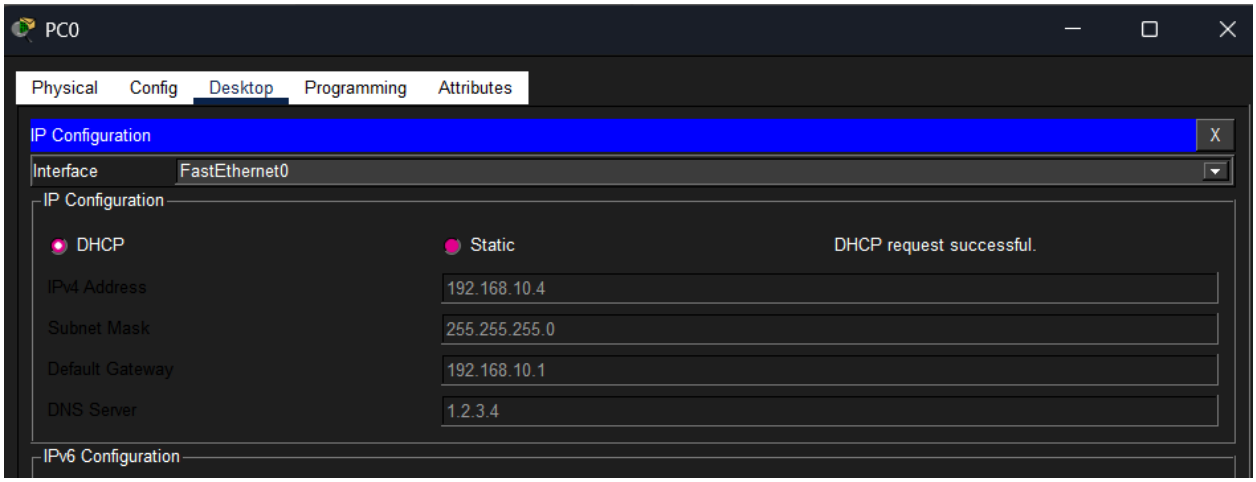


Router-

```
Router>enable
Router#config terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface gigabitethernet 0/0
Router(config-if)#ip address 192.168.10.1 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#ip dhcp pool xyz
Router(dhcp-config)#network 192.168.10.0 255.255.255.0
Router(dhcp-config)#default-router 192.168.10.1
Router(dhcp-config)#dns-server 1.2.3.4
Router(dhcp-config)#exit
```

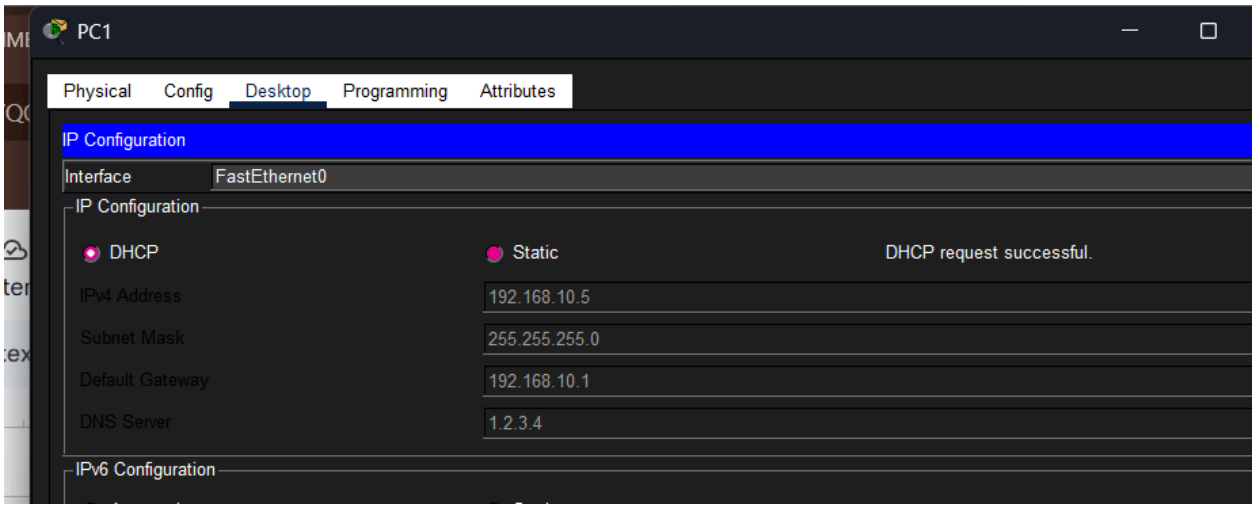
Pc0

-dhcp

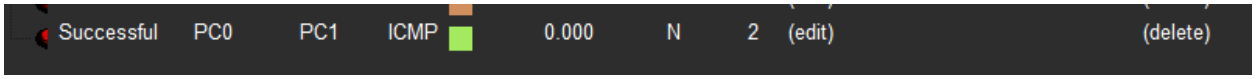


Pc1

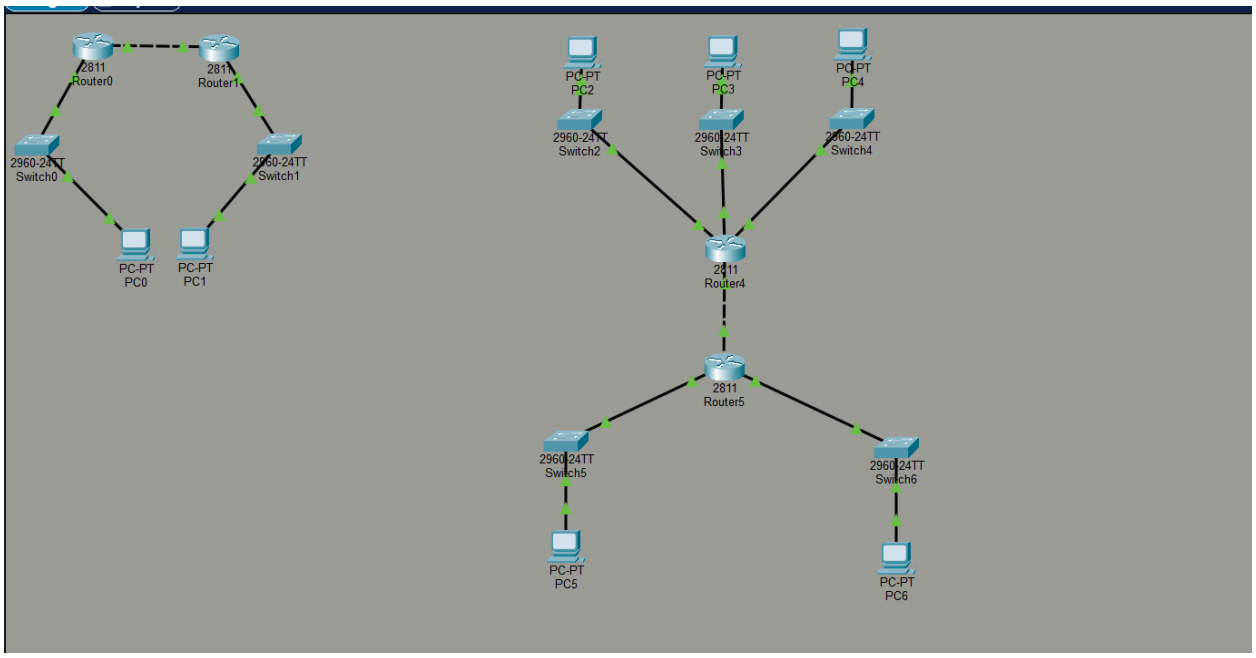
-dhcp



Output-



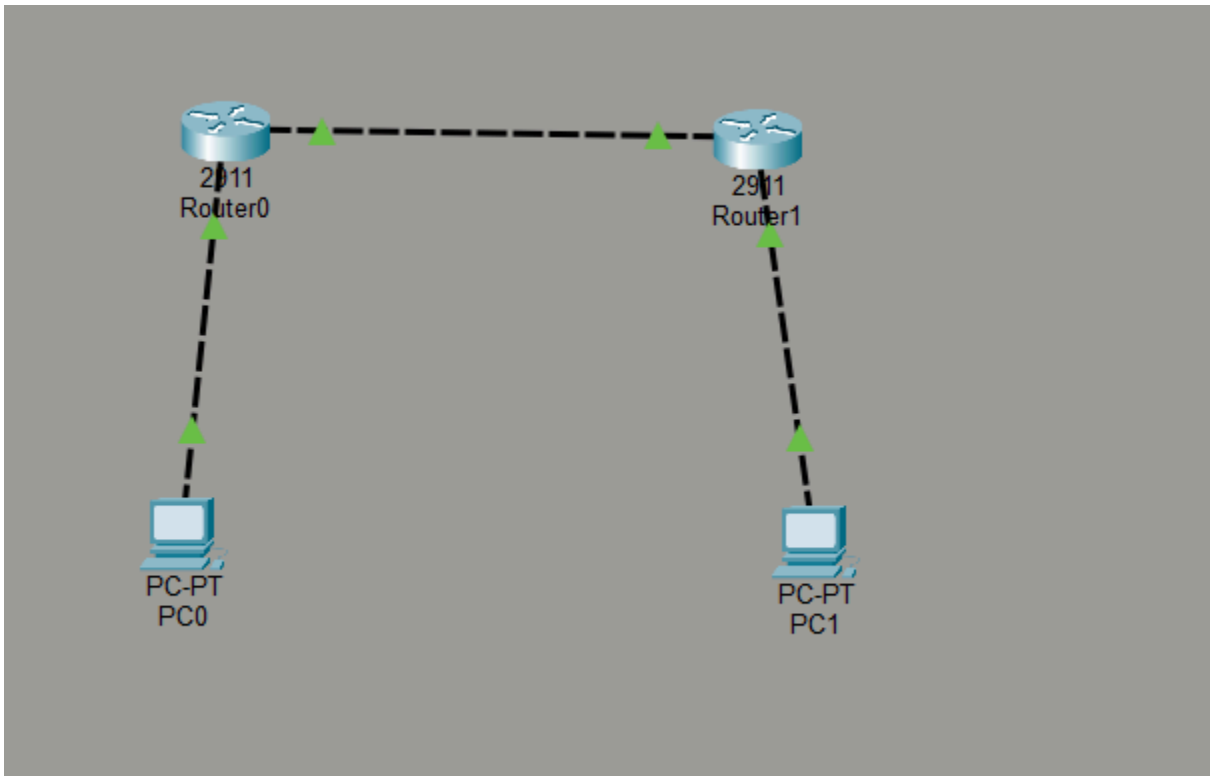
Assignment 5



Realtime Sim

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	PC5	PC6	ICMP		0.000	N	0	(edit)	(delete)
	Successful	PC0	PC1	ICMP		0.000	N	1	(edit)	(delete)

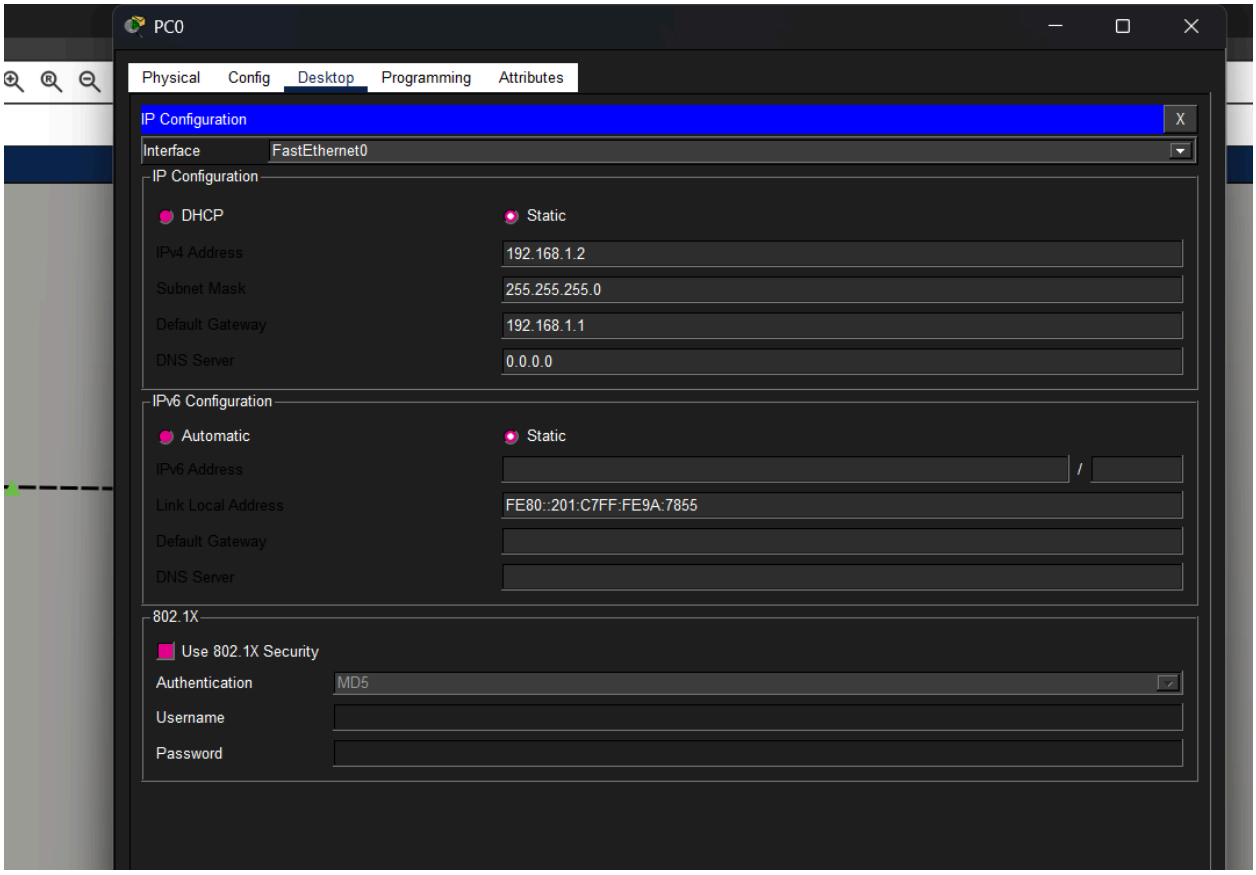
1-apr-2026- Example of 6th Assignment



```
Router>enable
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface gigabitethernet0/0
Router(config-if)#ip address 10.0.0.8
% Incomplete command.
Router(config-if)#ip address 10.0.0.8 255.0.0.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up
exit
Router(config)#interface gigabitethernet0/1
Router(config-if)#ip address 192.168.1.1 255.255.255.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
exit
Router(config)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
```

```
Router>
Router>enable
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#router rip
Router(config-router)#version 2
Router(config-router)#no autosummary
                             ^
% Invalid input detected at '^' marker.

Router(config-router)#
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#
Router#
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#router rip
Router(config-router)#no auto summary
Router(config-router)#network 192.168.1.0
Router(config-router)#network 10.0.0.0
Router(config-router)#exit
Router(config)#
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

Pc -0

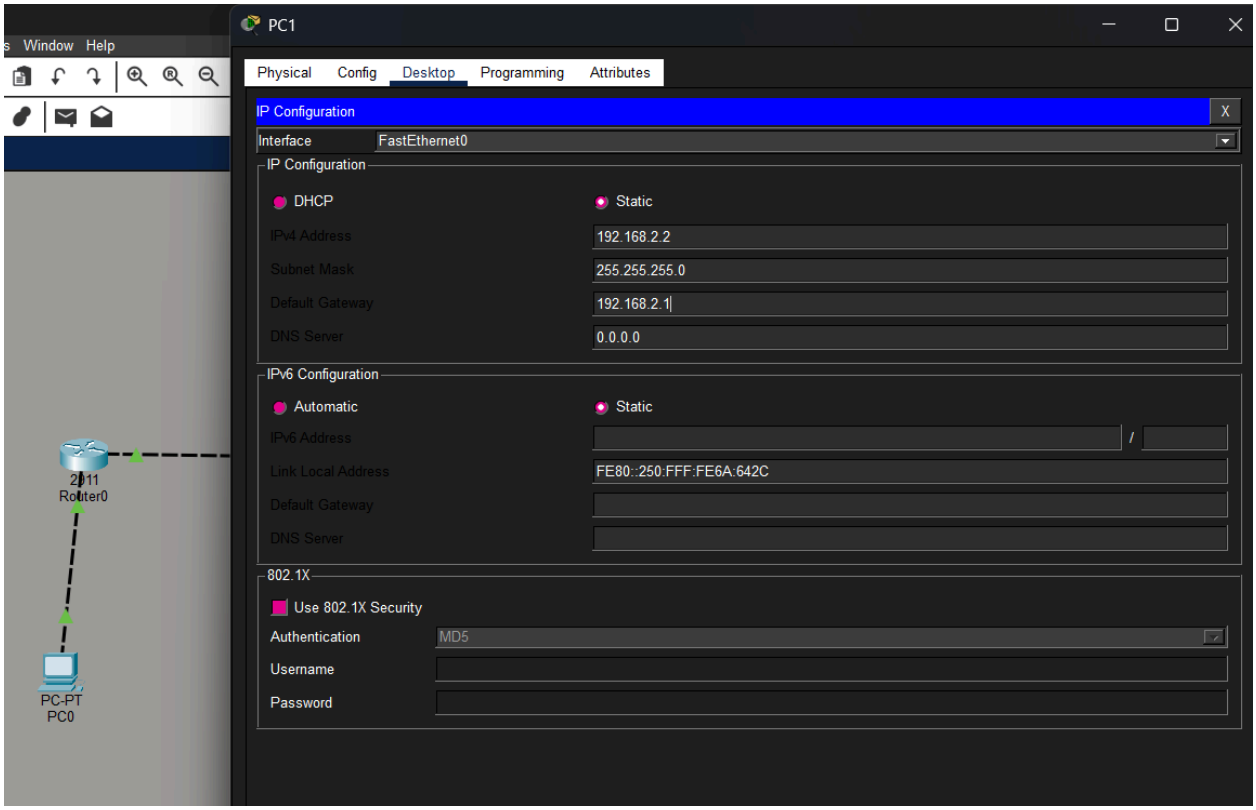

PC0
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.168.1.2
Subnet Mask
255.255.255.0
Default Gateway
192.168.1.1
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
/
Link Local Address
FE80::201:C7FF:FE9A:7855
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password

Router 2

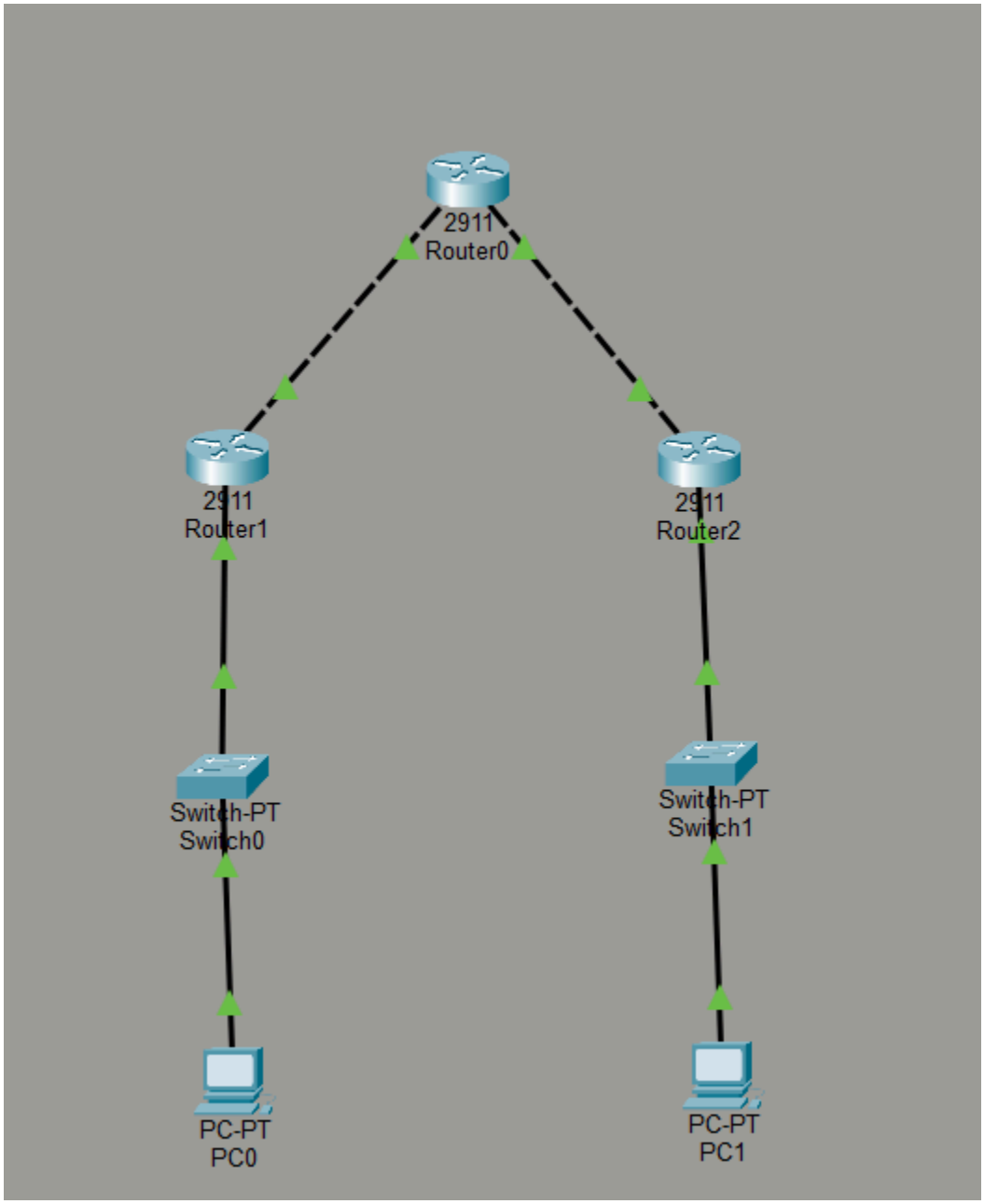
```
Router>enable
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface gigabitethernet0/0
Router(config-if)#ip address 10.0.0.9 255.255.255.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
exit
Router(config)#interface gigabitethernet0/1
Router(config-if)#ip address 192.168.2.1 255.255.255.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
exit
Router(config)#
```

```
Router>enable
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#router rip
Router(config-router)#version 2
Router(config-router)#no auto summary
Router(config-router)#network 192.168.2.0
Router(config-router)#network 10.0.0.0
Router(config-router)#exit
Router(config)#
Router(config)#
Router(config)#
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#enable
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#router rip
Router(config-router)#version 2
Router(config-router)#no auto summary
Router(config-router)#network 192.168.2.0
Router(config-router)#network 10.0.0.0
Router(config-router)#exit
Router(config)#
Router(config)#
Router(config)#
Router(config)#
Router(config)#
Router(config)#
Router(config)#
Router(config)#
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

Pc 1


Window Help
PC1
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.168.2.2
Subnet Mask
255.255.255.0
Default Gateway
192.168.2.1
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address
FE80::250:FFF:FE6A:642C
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password
2911
Router0
PC-PT
PC0

Assignment 6



Router 0-

```
Press RETURN to get started!



Router>enable
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface gig0/0
Router(config-if)#ip address 100.1.1.2 255.255.255.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

Router(config-if)#interface gig0/1
Router(config-if)#ip address 20.1.1.1 255.255.255.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

Router(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

Router(config-if)#router rip
Router(config-router)#version 2
Router(config-router)#no auto-summary
Router(config-router)#network 100.1.1.0
Router(config-router)#network 20.1.1.0
Router(config-router)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

Router 1-

```
Router>enable
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface gig0/0
Router(config-if)#ip address 100.1.1.1 255.255.255.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Router(config-if)#interface gig0/1
Router(config-if)#ip address 172.16.2.1 255.255.0.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

Router(config-if)#router rip
Router(config-router)#version 2
Router(config-router)#no auto-summary
Router(config-router)#network 100.1.1.0
Router(config-router)#network 172.16.0.0
Router(config-router)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

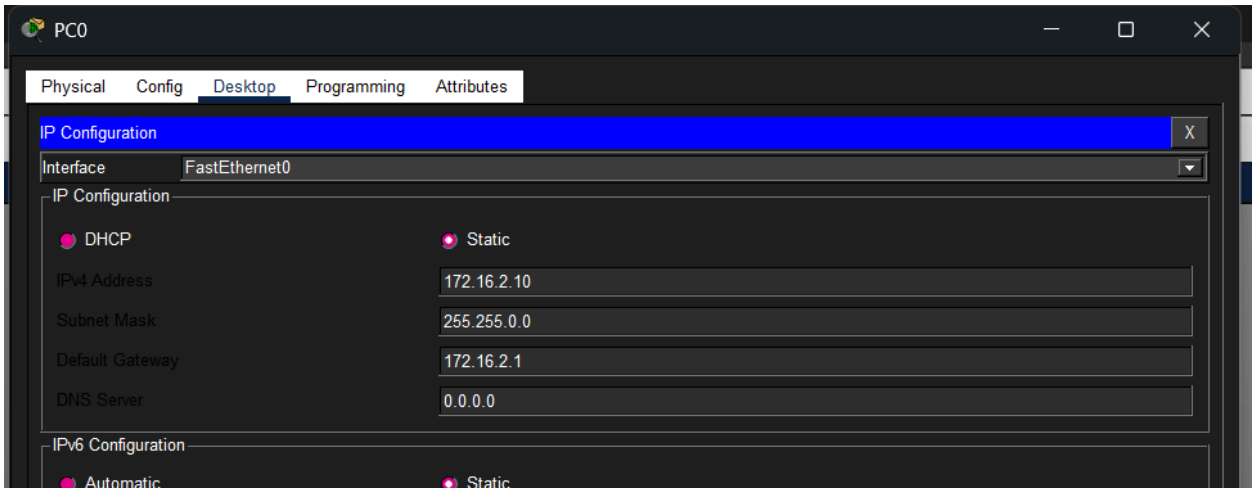
Copy

Router 2-

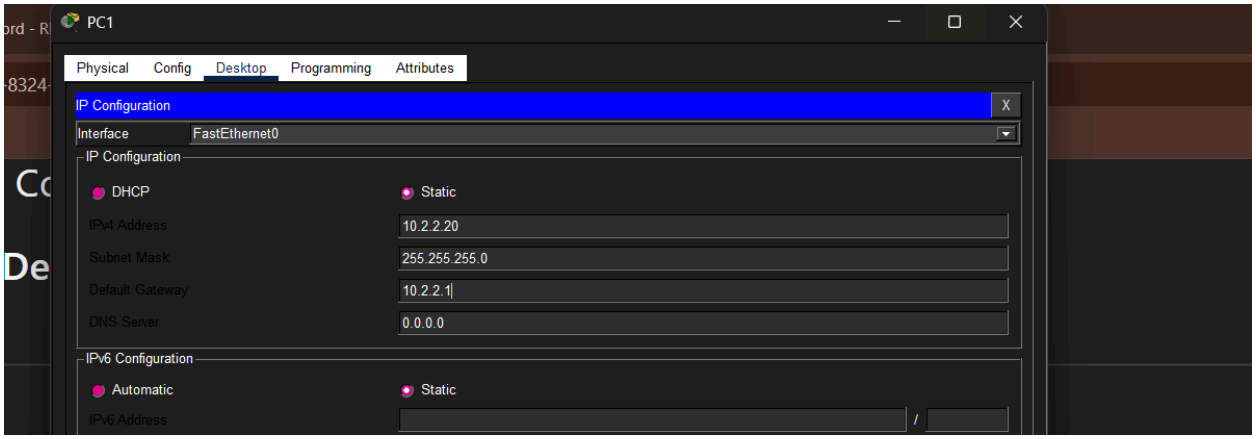
```
Router>enable
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface gig0/0
Router(config-if)#ip address 20.1.1.2 255.255.255.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Router(config-if)#interface gig0/1
Router(config-if)#ip address 10.2.2.1 255.255.255.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

Router(config-if)#router rip
Router(config-router)#version 2
Router(config-router)#no auto-summary
Router(config-router)#network id 20.1.1.0
                             ^
% Invalid input detected at '^' marker.

Router(config-router)#network 20.1.1.0
Router(config-router)#network 10.2.2.0
Router(config-router)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

Copy Paste

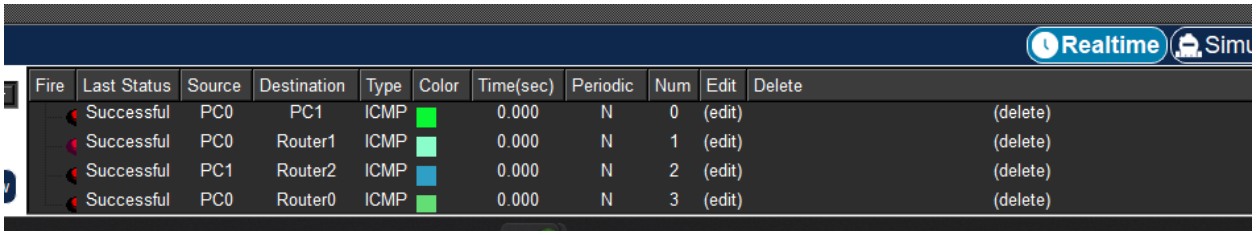
Pc 0-


PC0
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
172.16.2.10
Subnet Mask
255.255.0.0
Default Gateway
172.16.2.1
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static

Pc1-


PC1
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
10.2.2.20
Subnet Mask
255.255.255.0
Default Gateway
10.2.2.1
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address

o/p



Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	PC0	PC1	ICMP		0.000	N	0	(edit)	(delete)
	Successful	PC0	Router1	ICMP		0.000	N	1	(edit)	(delete)
	Successful	PC1	Router2	ICMP		0.000	N	2	(edit)	(delete)
	Successful	PC0	Router0	ICMP		0.000	N	3	(edit)	(delete)

15-04-26 - OSPF

WILD CARD MASK

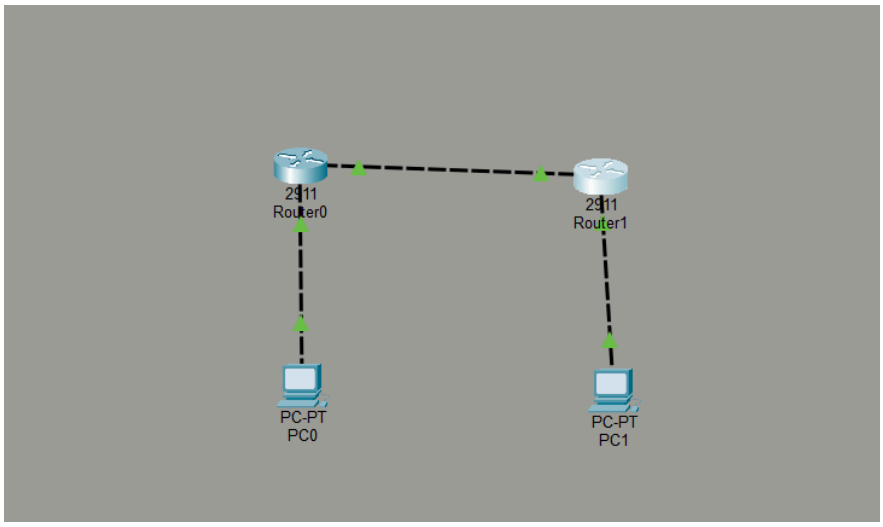
Network 192.168.5.0

If subnet = 255.255.255.0 , make 0 = 1 and 1 = 0

I.e = 0.0.0.255

10.1.1.0/30 = 255.255.252.0

For 10.1.1.0 , subnet = 255.255.252.0 i.e = (Wild card mask) 0.0.0.3



Router 0-

Router0

Physical | Config | CLI | Attributes

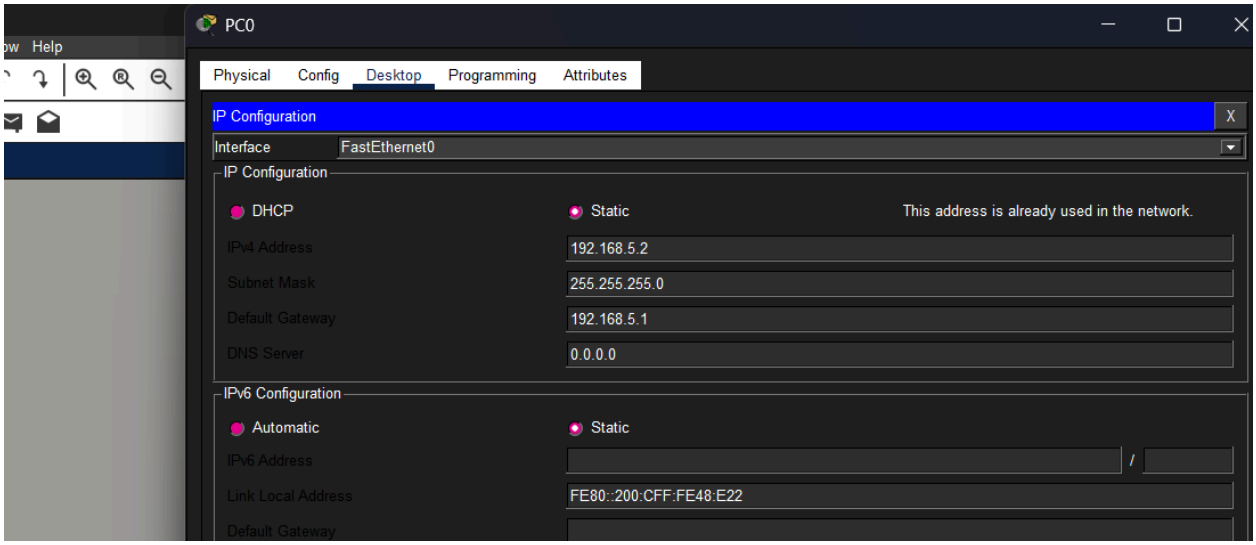
```
Router>enable
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#ip interface gig0/0
                  ^
% Invalid input detected at '^' marker.

Router(config)#interface 0/0
                         ^
% Invalid input detected at '^' marker.

Router(config)#config t
%Invalid hex value
Router(config)#interface gig0/0
Router(config-if)#ip address 10.1.1.1 255.255.252.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up
exit
Router(config)#config t
%Invalid hex value
Router(config)#interface gig0/1 192.168.5.1
                                ^
% Invalid input detected at '^' marker.

Router(config)#interface gig0/1
Router(config-if)#ip address 192.168.5.1 255.255.255.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
exit
Router(config)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
%IP-4-DUPADDR: Duplicate address 192.168.5.1 on GigabitEthernet0/1, sourced by 0000.0C48.0E22

Router(config)#config t
%Invalid hex value
Router(config)#router ospf 1
Router(config-router)#router-id 1.1.1.1
Router(config-router)#network 192.168.5.0
% Incomplete command.
Router(config-router)#network 192.168.5.0 0.0.0.255 area 0
Router(config-router)#network 10.1.1.0 0.0.0.3 area 0
Router(config-router)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

00:10:44: %OSPF-5-ADJCHG: Process 1, Nbr 2.2.2.2 on GigabitEthernet0/0 from LOADING to FULL, Loading Done
```

Copy | Paste

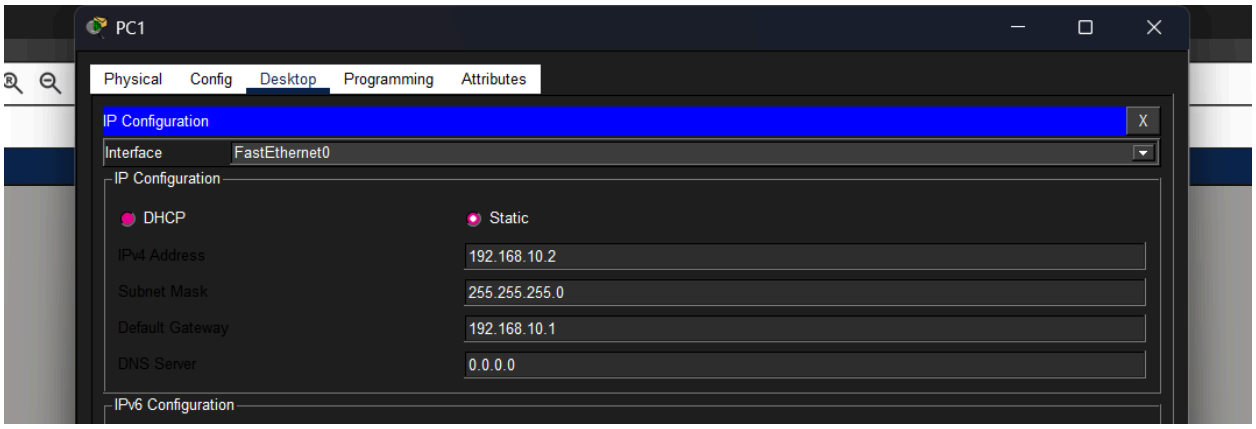
Top

Search

pc0-



Pc1



Router 1

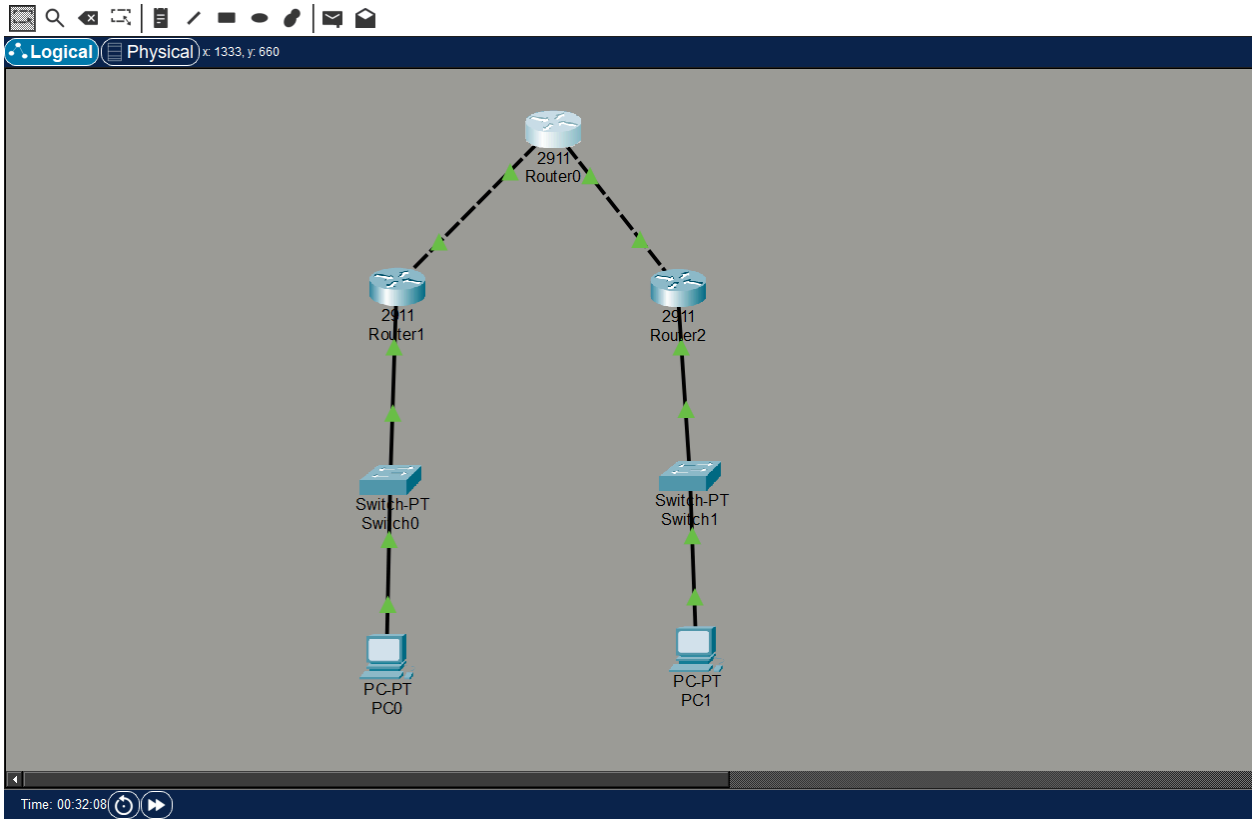
Router1

Physical Config CLI Attributes

```
         --- System Configuration Dialog ---

Would you like to enter the initial configuration dialog? [yes/no]: no


Press RETURN to get started!



Router>enable
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface gig0/0
Router(config-if)#ip address 10.1.1.2 255.255.252.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
exit
Router(config)#interface gig0/1
Router(config-if)#ip address 192.168.10.1
% Incomplete command.
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
exit
Router(config)#
Router(config)#config t
%Invalid hex value
Router(config)#router ospf 1
Router(config-router)#router 2.2.2.2
Router(config-router)#router-id 2.2.2.2
Router(config-router)#network id 192.168.10.0 0.0.0.255 area 0
                                 ^
% Invalid input detected at '^' marker.

Router(config-router)#network 192.168.10.0 0.0.0.255 area 0
Router(config-router)#network 10.1.1.0 0.0.0.3 area 0
Router(config-router)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

00:10:39: %OSPF-5-ADJCHG: Process 1, Nbr 1.1.1.1 on GigabitEthernet0/0 from LOADING to FULL, Loading Done

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
```

Copy Paste

ASSIGN -7


Logical
Physical
x: 1333, y: 660
2911
Router0
2911
Router1
2911
Router2
Switch-PT
Switch0
Switch-PT
Switch1
PC-PT
PC0
PC-PT
PC1
Time: 00:32:08

Router 0

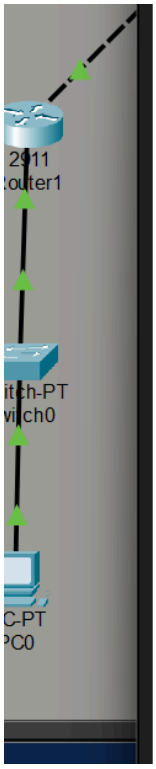
Ip ADD - 100.1.1.0 below not 10.1.1.0

```
Router>enable
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface gig0/0
Router(config-if)#ip address 10.1.1.0 255.255.255.0
Bad mask /24 for address 10.1.1.0
Router(config-if)#ip address 10.1.1.1 255.255.255.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up
exit
Router(config)#config t
%Invalid hex value
Router(config)#interface gig0/1
Router(config-if)#ip address 20.1.1.1 255.255.255.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up
exit
Router(config)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

Router(config)#config t
%Invalid hex value
Router(config)#router ospf 1
Router(config-router)#router-id 1.1.1.1
Router(config-router)#100.1.1.0 0.0.0.255 area 0
                     ^
% Invalid input detected at '^' marker.

Router(config-router)#network 100.1.1.0 0.0.0.255 area 0
Router(config-router)#network 20.1.1.0 0.0.0.255 area 0
Router(config-router)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

00:16:11: %OSPF-5-ADJCHG: Process 1, Nbr 3.3.3.3 on GigabitEthernet0/1 from LOADING to FULL, Loading Done
```

Router 1

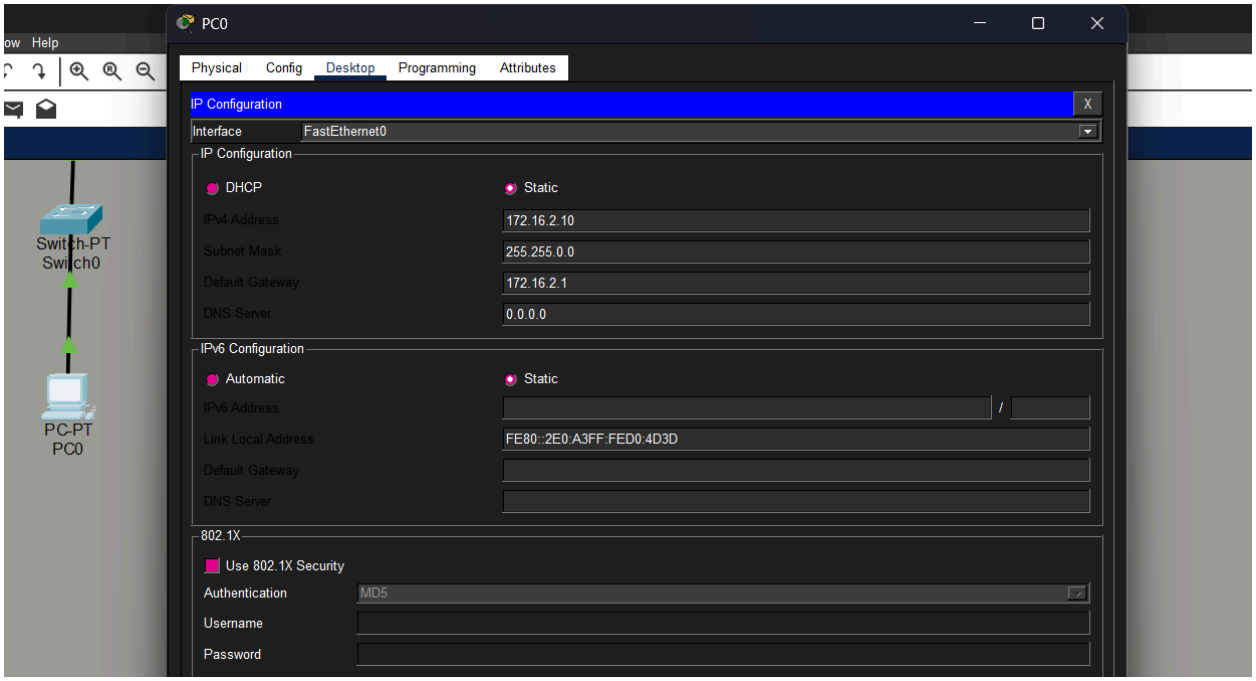
```
Router>
Router>
Router>enable
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface gig0/0
Router(config-if)#ip address 100.1.1.2 255.255.255.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
exit
Router(config)#
% Unknown command or computer name, or unable to find computer address

Router(config)#config t
%Invalid hex value
Router(config)#interface gig0/1
Router(config-if)#ip address 172.16.2.1 255.255.0.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
exit
Router(config)#config t
%Invalid hex value
Router(config)#router-id 2.2.2.2
                        ^
% Invalid input detected at '^' marker.

Router(config)#router ospf 1
Router(config-router)#router-id 2.2.2.2
Router(config-router)#network 100.1.1.0 0.0.0.255 area 0
Router(config-router)#network 172.16.0.0 0.0.255.255 area 0
Router(config-router)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

Router 2



```
Router>enable
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface gig0/0
Router(config-if)#ip address 20.1.1.2 255.255.255.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
exit
Router(config)#interface gig0/1
Router(config-if)#ip address 10.2.2.1 255.255.255.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
exit
Router(config)#router ospf 1
Router(config-router)#router-id 3.3.3.3
Router(config-router)#network-id 20.1.1.0 0.0.0.255 area 0
                                         ^
% Invalid input detected at '^' marker.

Router(config-router)#network 20.1.1.0 0.0.0.255 area 0
Router(config-router)#network 10.2.2.0 0.0
00:16:05: %OSPF-5-ADJCHG: Process 1, Nbr 1.1.1.1 on GigabitEthernet0/0 from LOADING to FULL, Loading Done
.0
                                         ^
% Invalid input detected at '^' marker.

Router(config-router)#network 10.2.2.0 0.0.0.255 area 0
Router(config-router)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

PC0-

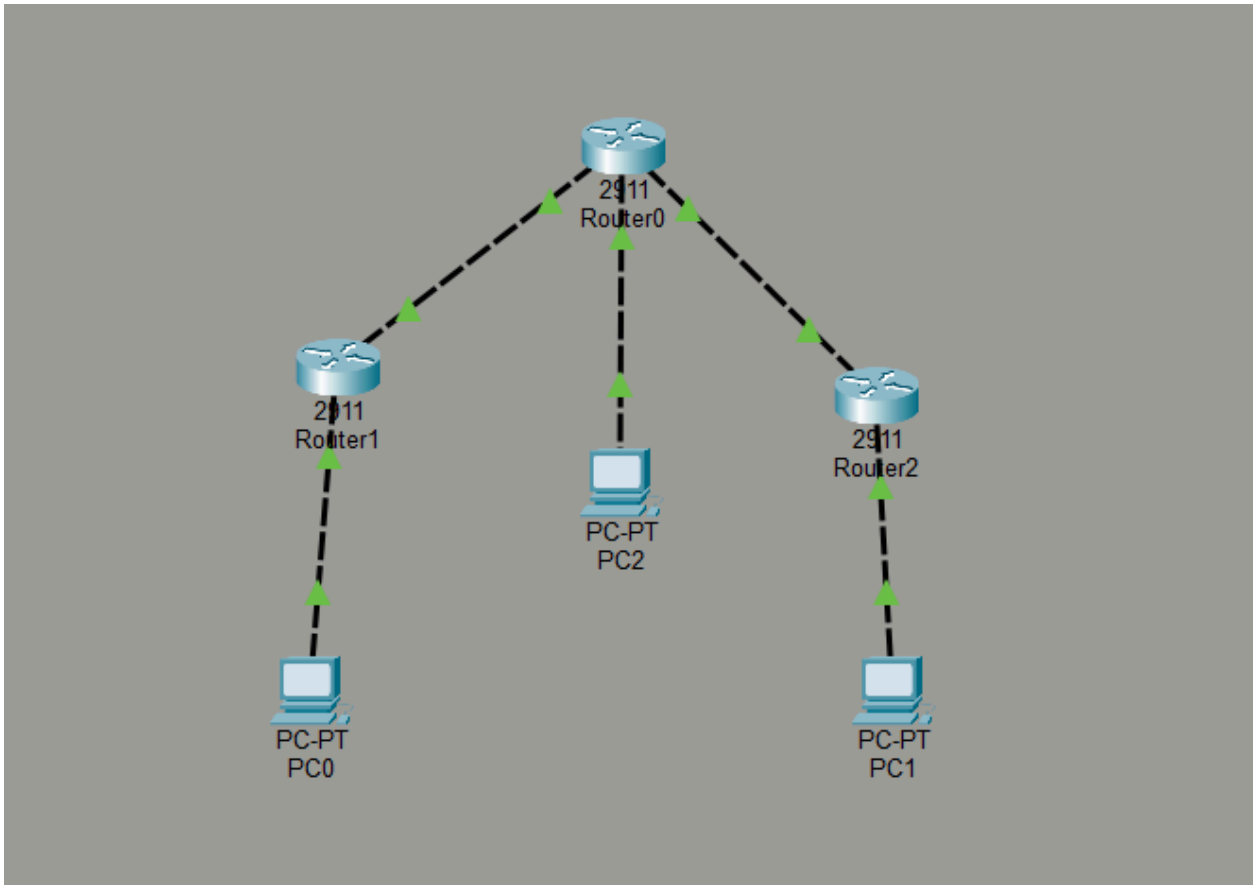


PC1-

PC1
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
10.2.2.20
Subnet Mask
255.0.0.0
Default Gateway
10.2.2.1
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static

ASSIGNMENT 8

Example -



Router -0

```
Router>
Router>enable
Router#interface gig0/0
                    ^
% Invalid input detected at '^' marker.

Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface gig0/0
Router(config-if)#ip address 172.16.0.2
% Incomplete command.
Router(config-if)#ip address 172.16.0.2 255.255.0.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up
exit
Router(config)#config t
%Invalid hex value
Router(config)#interface gig0/1
Router(config-if)#ip address 172.14.0.1 255.255.0.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up
exit
Router(config)#config t
%Invalid hex value
Router(config)#ip address 50.0.0.1 255.0.0.0
                    ^
% Invalid input detected at '^' marker.

Router(config)#interface gig0/2
Router(config-if)#ip address 50.0.0.1 255.0.0.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/2, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to up
exit
Router(config)#
```

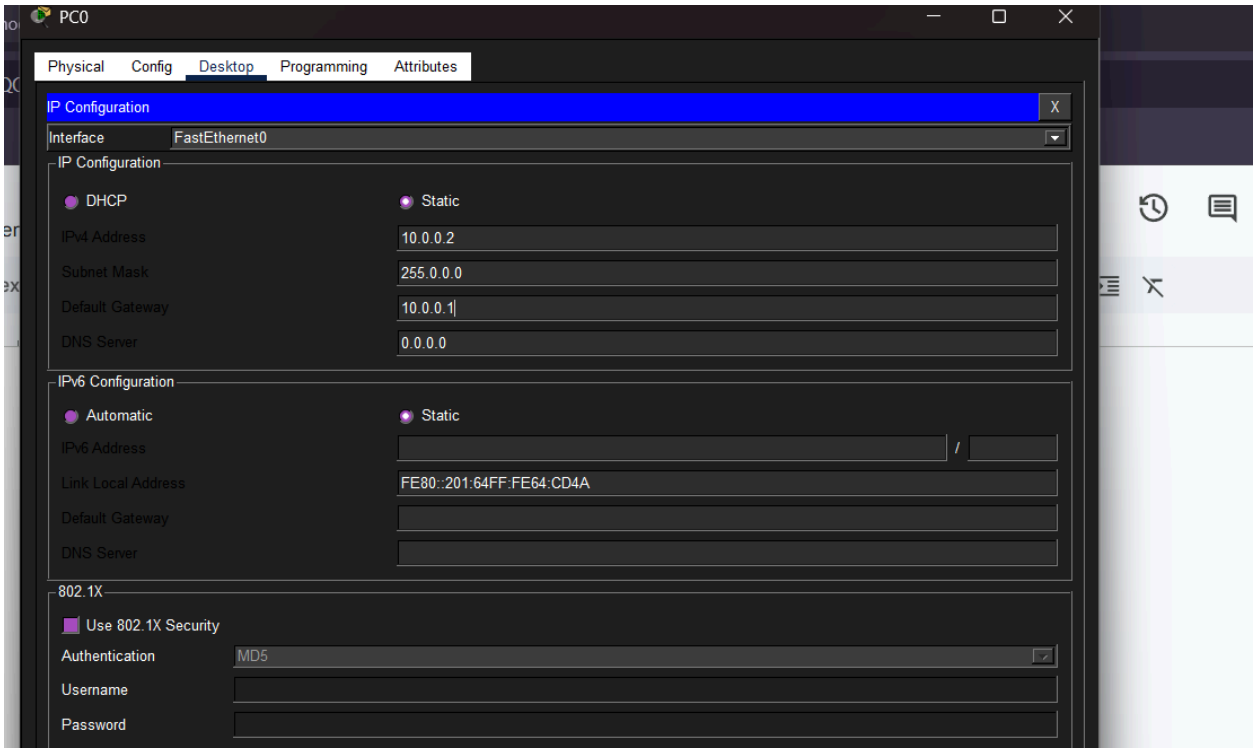
Router 1

```
Router>enable
Router#conig t
                ^
% Invalid input detected at '^' marker.

Router#
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface gig0/0
Router(config-if)#ip address 172.16.0.1 255.255.0.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
exit'
                      ^
% Invalid input detected at '^' marker.

Router(config-if)#exit
Router(config)#config t
%Invalid hex value
Router(config)#interface gig0/1
Router(config-if)#ip address 10.0.0.1 255.0.0.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
exit
Router(config)#
```

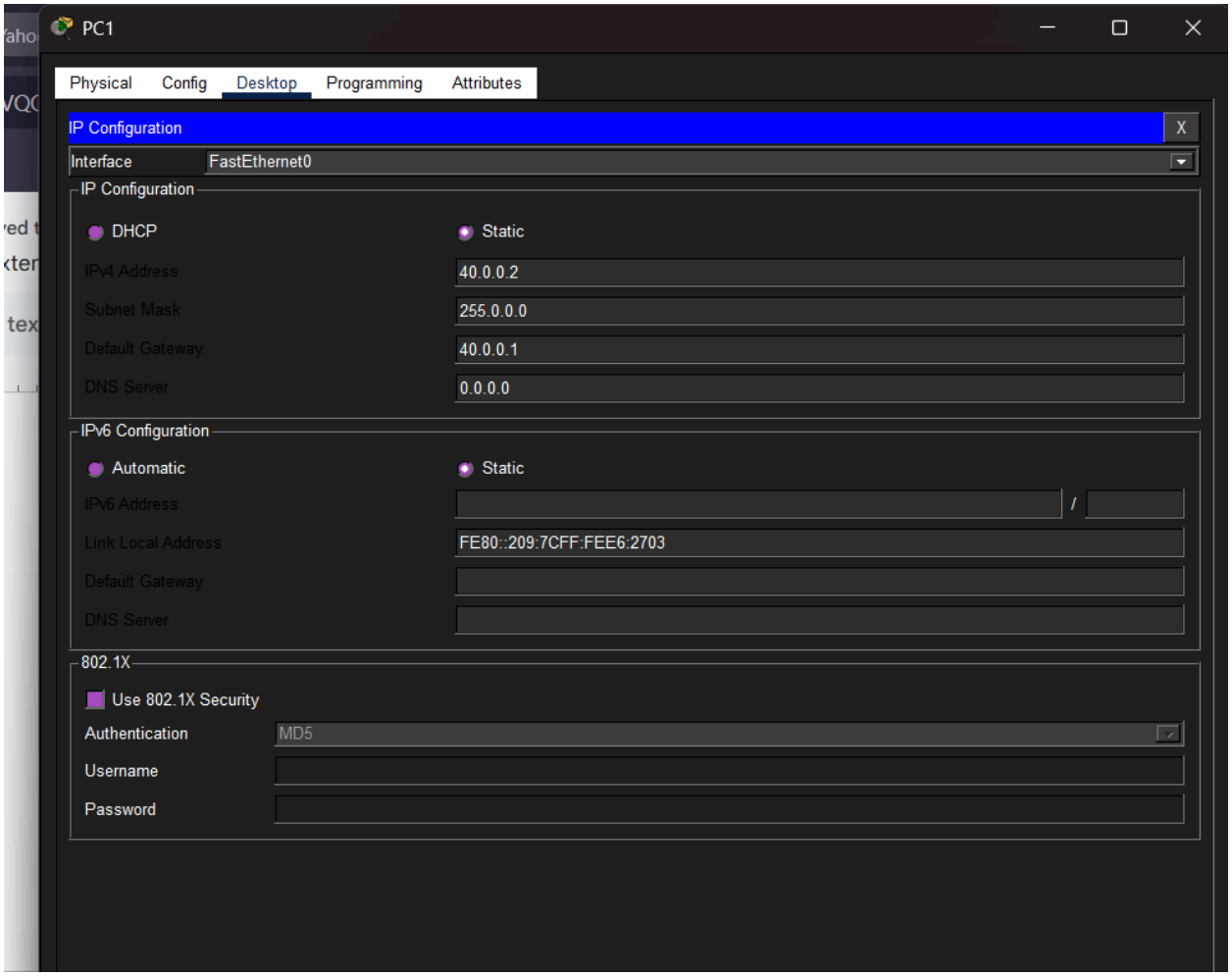
Router 2

```
Router>enable
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface gig0/0
Router(config-if)#ip address 172.14.0.2 255.255.0.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
exit
Router(config)#
Router(config)#config t
%Invalid hex value
Router(config)#interface gig0/1
Router(config-if)#ip address 40.0.0.1 255.0.0.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
exit
Router(config)#
```

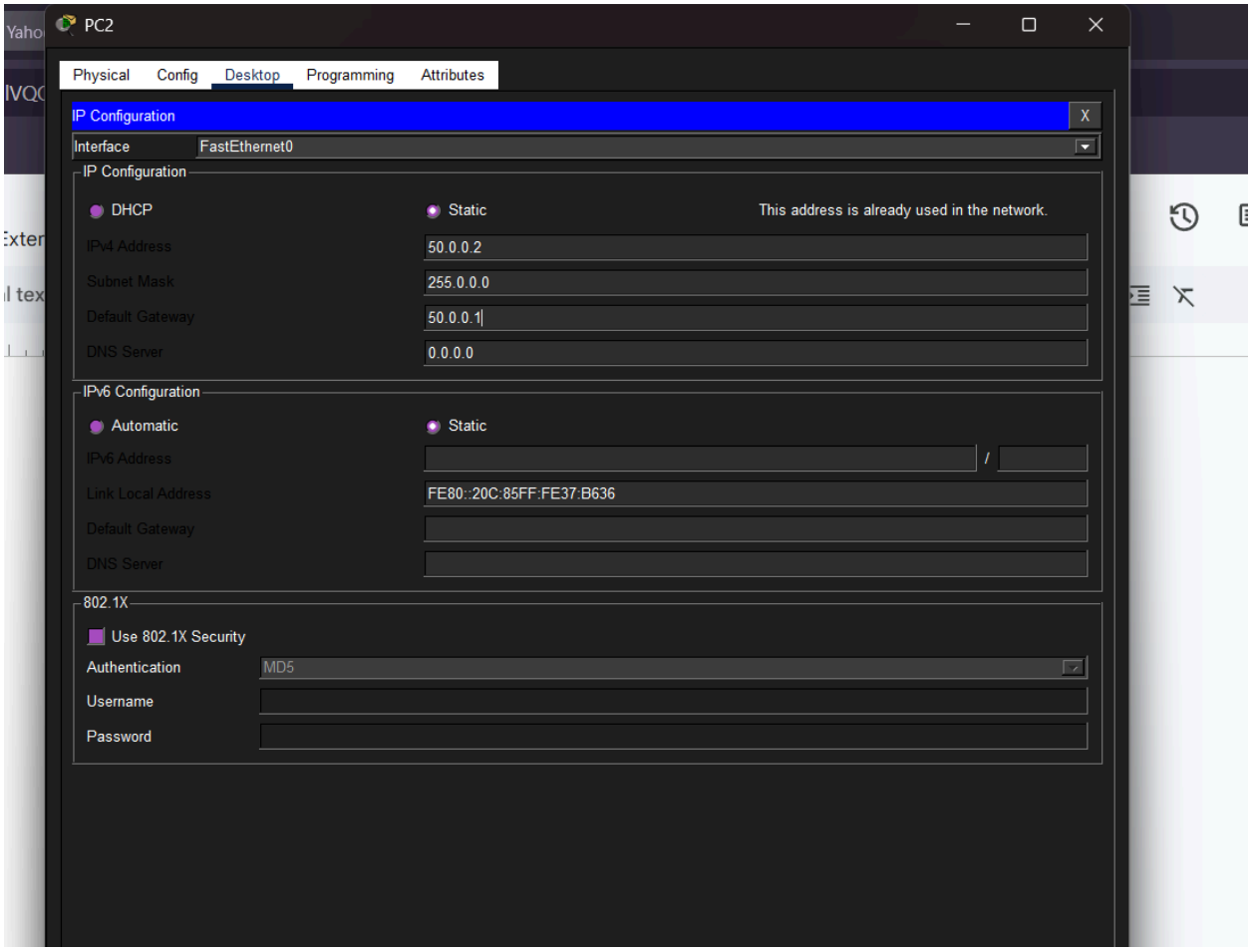
Pc0



Pc1


PC1
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
40.0.0.2
Subnet Mask
255.0.0.0
Default Gateway
40.0.0.1
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
/
Link Local Address
FE80::209:7CFF:FEE6:2703
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password

Pc2-



BGP

Router 0 -cont..

```
Router(config)#router bgp 71
BGP is already running; AS is 1
Router(config)#no router bgp 1
Router(config)#router bgp 71
Router(config-router)#neighbor 172.16.0.1 remote as 1
                                                  ^
% Invalid input detected at '^' marker.

Router(config-router)#neighbor 172.16.0.1 remote-as 1
Router(config-router)#neighbor 172.14.0.2 remote-as 79
Router(config-router)#network 50.0.0.0 mask 255.0.0.0
Router(config-router)#exit
Router(config)#%BGP-5-ADJCHANGE: neighbor 172.16.0.1 Up
%BGP-5-ADJCHANGE: neighbor 172.14.0.2 Up
```

Router 1-cont..

```
Router>
Router>enable
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#router bgp 1
Router(config-router)#neighbor 172.16.0.2 remote-as 71
Router(config-router)#%BGP-5-ADJCHANGE: neighbor 172.16.0.2 Up
network
% Incomplete command.
Router(config-router)#network 10.0.0.0 mask 255.0.0.0
Router(config-router)#exit
Router(config)#
```

Copy

Router 2-cont..

```
Router>
Router>enable
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#router bgp 79
Router(config-router)#neighbor 172.14.0.1 remote-as 71
Router(config-router)#%BGP-5-ADJCHANGE: neighbor 172.14.0.1 Up

Router(config-router)#network 40.0.0.0 mask 255.0.0.0
Router(config-router)#exit
Router(config)#
```

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Scenario 0

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Toggle PDU List Window

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	PC2	Router0	ICMP		0.000	N	2	(edit)	(delete)
	Successful	PC0	PC2	ICMP		0.000	N	3	(edit)	(delete)
	Successful	PC0	PC1	ICMP		0.000	N	4	(edit)	(delete)